# Aras Technical Documentation 14

## User's Guide

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# Copyright Information



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# Document Conventions

The following table highlights the document conventions used in the document:

Table 1: Document Conventions

| Convention | Description |
| --- | --- |
| **Bold** | This shows the names of menu items, dialog boxes, dialog box elements, and commands.<br>Example: Click **OK**. |
| `Code` | Code examples appear in `courier` font. It may represent text you type or data you read. |
| `Yellow highlight` | Code highlighted in yellow draws attention to the code that is being indicated in the content. |
| `Yellow highlight with red text` | Red text highlighted in yellow indicates the code parameter that needs to be changed or replaced. |
| *Italics* | Reference to other documents. |
| **Note:** | Notes contain additional useful information. |
| **Warning** | Warnings contain important information. Pay special attention to information highlighted this way. |
| Successive menu choices | Successive menu choices may appear with a greater than sign (-->) between the items that you will select consecutively.<br>Example: Navigate to **File** --> **Save** --> **OK**. |

# 1 Overview

The Aras Technical Documentation Application (Tech Docs) is a content authoring tool, fully integrated into Aras Innovator PLM that collectively provides component content management capability for topics-based, modular documentation. Tech Docs enables users of various disciplines to author, visualize, share, and publish information all within a central, controlled, and collaborative environment. Unlike other systems that provide Document-level Management functions for content produced by external applications, Aras Tech Docs leverages Innovator's core PLM functionality (Access Control, Workflow/Lifecycle, Change Notification, etc.) and provides a web-based authoring tool for creating content directly, embedding shared document components, or referencing other elements managed by the PLM system. So as related Documents, Parts, Project Tasks, Requirements, or any other information managed by Aras Innovator evolve through subsequent revisions, authors of technical documentation can be alerted and navigated to pending changes, or the system can update the related documentation automatically.

Technical Documents can be composed of individual components (sections, paragraphs, illustrations, tables, etc.), where each component can change independently; and optionally created/managed by multiple users. The aggregated information forms the complete published document. This Structured Content is specified by a user-defined data model that identifies the elements of each component, their relationships and cardinality, and any restrictions on data type and value limitations. The built-in authoring User Interface directs the author as to the types of document elements that can be inserted, constantly validating the document as it is created. Candidate documents for release can be routed through workflow and published on final acceptance.

This document describes the functions and features of the Technical Documentation application. It is meant to be a reference for the Technical Document Author (Section 5.2.1) as they use the software to author technical content. Section 1 provides an overview of the features of the software. Section 2 describes the User Interface. Section 3 describes how you use the software to create technical content. Section 4 describes how to publish/export content in various formats. Section 5 discusses how Technical Documents are managed in the Innovator environment.

## 1.1 Terms

The following Terms are used throughout this document:

| Term | Definition |
|---|---|
| Business Object | Refers to any Item in Innovator that contains information, which may be referred to in a Technical Document. |
| Cascading Style Sheets (CSS) | Industry standard text-based format for controlling the position, color, size, etc. of web-based content. It is used by the Technical Documentation application to control the format of HTML and PDF content. |
| Content Filters | Metadata (name/value pairs) that are defined by the Technical Document Administrator and chosen by the Technical Document Author to characterize Document Elements within a Technical Document. |
| Context Menus | Pop-up menus, used mostly by Document Elements, that contain menu items used to invoke various functions on the Document Element instance. The term 'context' is used because the specific menu items displayed in the menu may be different depending on the specific Document Element instance selected. |
| Document Element | Refers to the components that make up a Technical Document. Every Document Element is associated with a specific node in the Navigation Pane and XML Element in the resulting content. Document Elements are explicitly defined in the Technical Document Type Definition. |
| Document Style | Refers to the formatting and positional logic that defines the placement and look-and-feel of the rendered content of a Technical Document. Document Style is controlled using Cascading Style Sheets (CSS), associated with a specific Document Type, and defined by the Technical Document Administrator. |
| Graphic | Any 2D image that can be displayed in a Technical Document. |
| Mapped Property | Refers to a Document Element with content that is automatically created from a Property of a referenced Item |
| Structured Content | Refers to the general use of an underlying schema to define the content of a document. Structured Content is contrasted with unstructured (or freeform) content in which case there are no rules that help define and control the content. |
| Technical Document | Refers to instances of a Technical Document ItemType. In this case, the term is capitalized. When lower case, the context is more general – any technical document. |
| Technical Document Type | Refers to instances of a Document Type Item, which is used to configure the Structured Content and Document Style for a Technical Document. |

# 1.2 Features

This section provides a brief overview of each of the main features of the Technical Documentation application with links to associated sections for more information.

## 1.2.1 Structured Documents and Modules

Technical Documents can exist either as a single entity - with all content contained within – or aggregate content from multiple document *components*. Breaking up documents into smaller components has multiple advantages including the ability to:

1. Reuse each component.
2. Enable concurrent editing.
3. Apportion each component to areas of specific specialty or expertise

In the Technical Publication domain, technical content is sometimes referred to as 'Topics-based' or 'Modular'. This refers to content that is about a specific topic or area of focus and is, to some degree, context insensitive. This helps to characterize the content and enable it to exist in multiple places within a document. Contrast this with a narrative in other forms of documents where every paragraph builds upon (and thus depends on) previous paragraphs and context.

Each Technical Document instance can evolve independently and be protected by specific access rights. To be reusable, each document component needs to be constructed in a way that allows it to fit within a larger context. To achieve this, the content of each Document / Document Component will be based on an overarching schema or document structure (Section 3.1). This document structure helps ensure consistency and provides a mechanism for determining how individual components can fit into a larger context.

## 1.2.2 Document Content Styling

The style and layout of Technical Documents is dictated by a separate style configuration that is maintained separately but associated directly with a Technical Document Type definition. Each Document Element defined in the Schema can be referenced and have a unique set of styling parameters that are used to position and render the associated Document Element instances in each document. Style settings can be assigned to HTML output, PDF Output, and the Document Editor.

Referencing centrally controlled style settings ensure consistency in look and feel for published content and provides a convenient mechanism to update style in one location and have it automatically apply to all associated documents.

## 1.2.3 Metadata

All Document Element content can have additional metadata information assigned to provide semantics for content classification and aid in content search and retrieval. This metadata is provided by the Technical Document Author when the Document Structure is designed and configured.

### 1.2.4 Item Referencing

Technical Documents can have explicit relationships created with other Items maintained in Innovator – Business Objects (Section 3.2.7). This relationship establishes a direct binding between the document and associated Business Objects. In addition, property data from these Business Objects can be queried and used within the Technical Document (Section 3.7). When the Business Object property data is updated, the associated Technical Documents are updated accordingly. This binding helps maintain data integrity and identify a hard link across Objects and the documentation that was created specifically for them.

### 1.2.5 Conditional Content and Views

Technical documentation content should evolve as the information it references evolve. Typically, product information does not change substantially between versions. As a result, documentation about each version should conserve and reuse all that is common and identify the areas that are unique to one or more product versions. The Technical Documentation application provides a mechanism to tag elements within a document with one or more characteristics, a particular version or model number for example. These characteristics can then be used to filter (Section 3.3) document content to only display or include information that is relevant.

### 1.2.6 Publication and Data Export

The Technical Documentation application, together with Aras Innovator, provide an authoring and data management environment. To make use of the content created, it needs to be exposed outside the PLM environment. The publishing/export function (Section 4) provides an ability to convert the content of each Technical Document to a format more suitable to end-user consumption.

# 2 User Interface

Authoring and viewing Technical Document content is done using the Technical Document Form.

The Tear Off window for Items of type Technical Document has two views: Form and Editor. These Views are accessed using a sidebar menu displayed in the left margin.

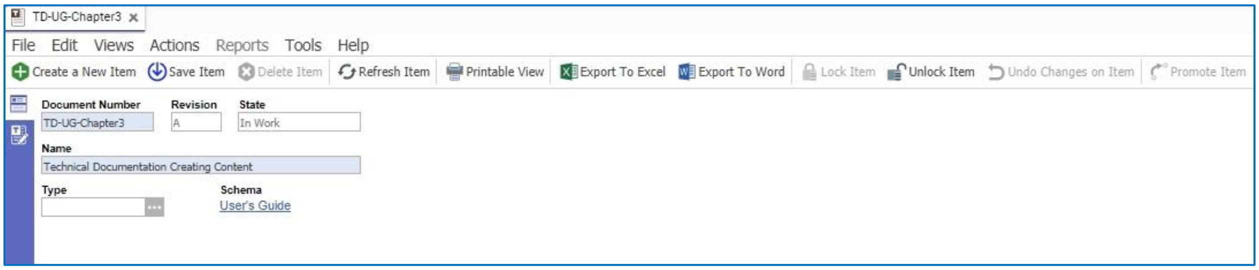



Figure 1.

Click the Form Button in the sidebar menu to display the Form. The Form shows all the properties of the Technical Document that have been exposed. If there are any related Items added to the Technical Document, they appear as separate tabs in the lower portion of the window as is usual for ItemType Forms.[1]

Click the Editor Button in the sidebar menu to display the Technical Document Editor.

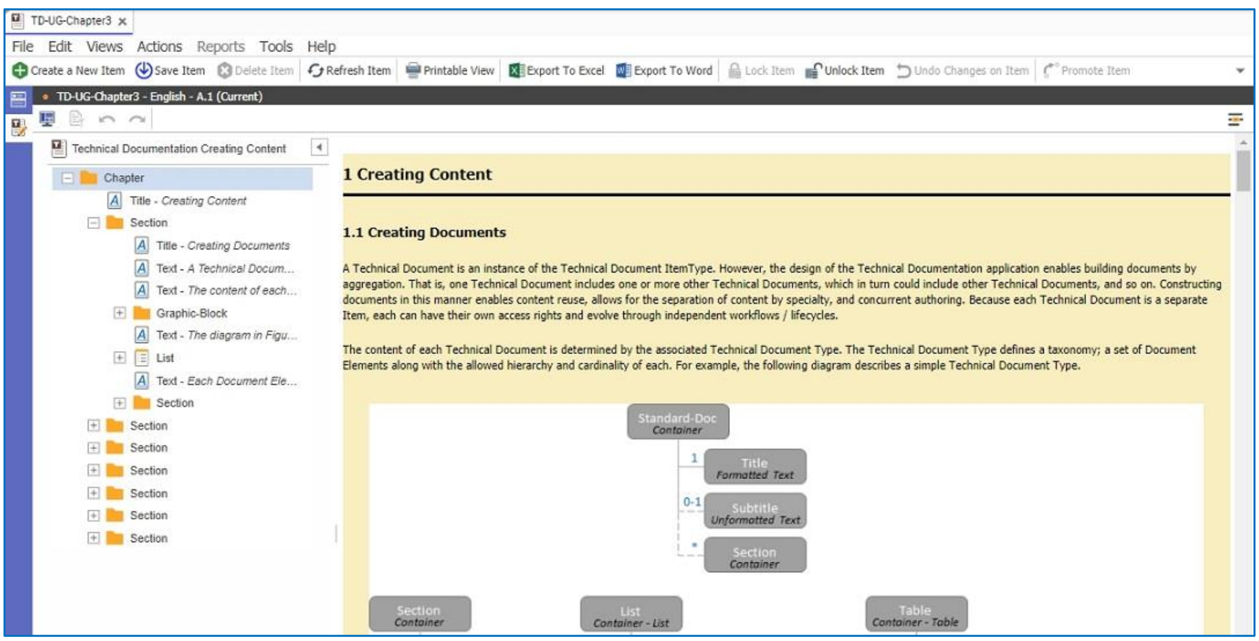



Figure 2.

[1] Relationship Item Tabs are turned off by default. Update the tp_Block ItemType to display Relationship tabs as required.

The Technical Document Editor contains three main panes: Toolbar, Navigation Pane, and the Content Viewer/Editor. The toolbar contains various buttons to invoke separate functions for viewing and content editing. The buttons exposed on the toolbar are context-sensitive based on the currently selected Document Element. The navigation pane (sometimes called Structure Tree) displays individual 'nodes' each representing the hierarchy of Document Elements that comprise the document. Selecting the '+' icon at various levels expands the tree by displaying all the *child* Document Elements. Selecting a node in the navigation pane causes the corresponding content associated with that node to be scrolled and highlighted in the Content Viewer/Editor. The Content Viewer/Editor displays the content of the document and represents the canvas on which content is added and modified by the author. Selecting content in the editor/viewer causes the corresponding Document Element node in the navigation pane to be scrolled and highlighted.

## 2.1 Searching for Text

You can search for words or phrases within the document by clicking the icon that appears in the upper right corner of the document window. The icon is replaced by a search box. Enter your search term in the box and press Enter. If the search term you specified is contained in the document, all instances of the term appear in a different font color.

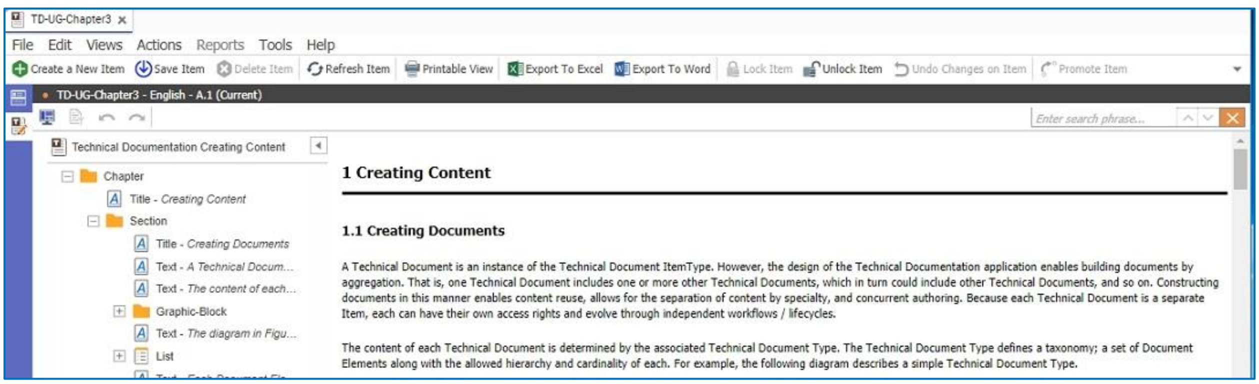



Figure 3.

You can scroll through the document by clicking on the up/down arrows in the Search box. The currently active match is highlighted as you scroll. The search box also displays the ordered position of the highlighted search term. If the document does not contain the search term you specified, the phrase "no matches are found" is displayed.

## 2.2 Replacing Text

To replace text, select the down arrow in the upper right corner of the Search/Replace dialog. Doing so will expose the *Replace* portion of the dialog. Select 'Replace Next' icon to step through the searched text results and replace with the text entered in the 'Replace' text box. Select the Replace All icon to replace all found search strings in the document. Note that only editable portions of the document can have the text updated.

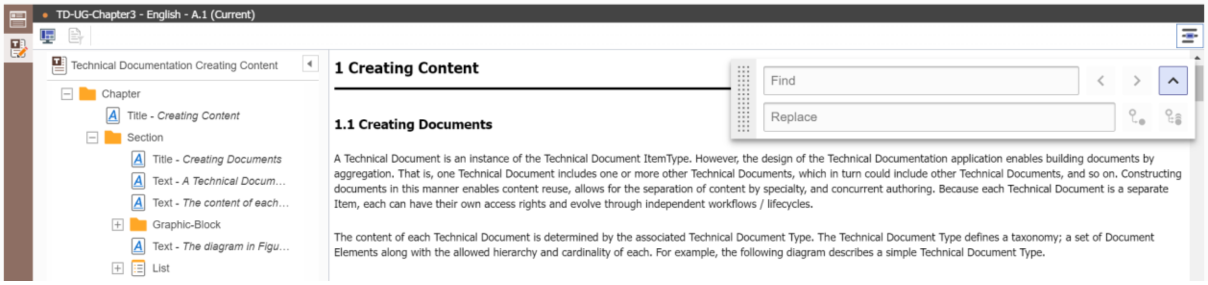



Figure 4.

# 2.3 Viewing Documents Side-by-Side

You can view different versions of the same document simultaneously using the following procedure:

1. Click the Document View icon  in the toolbar. The Document View dialogue box appears.

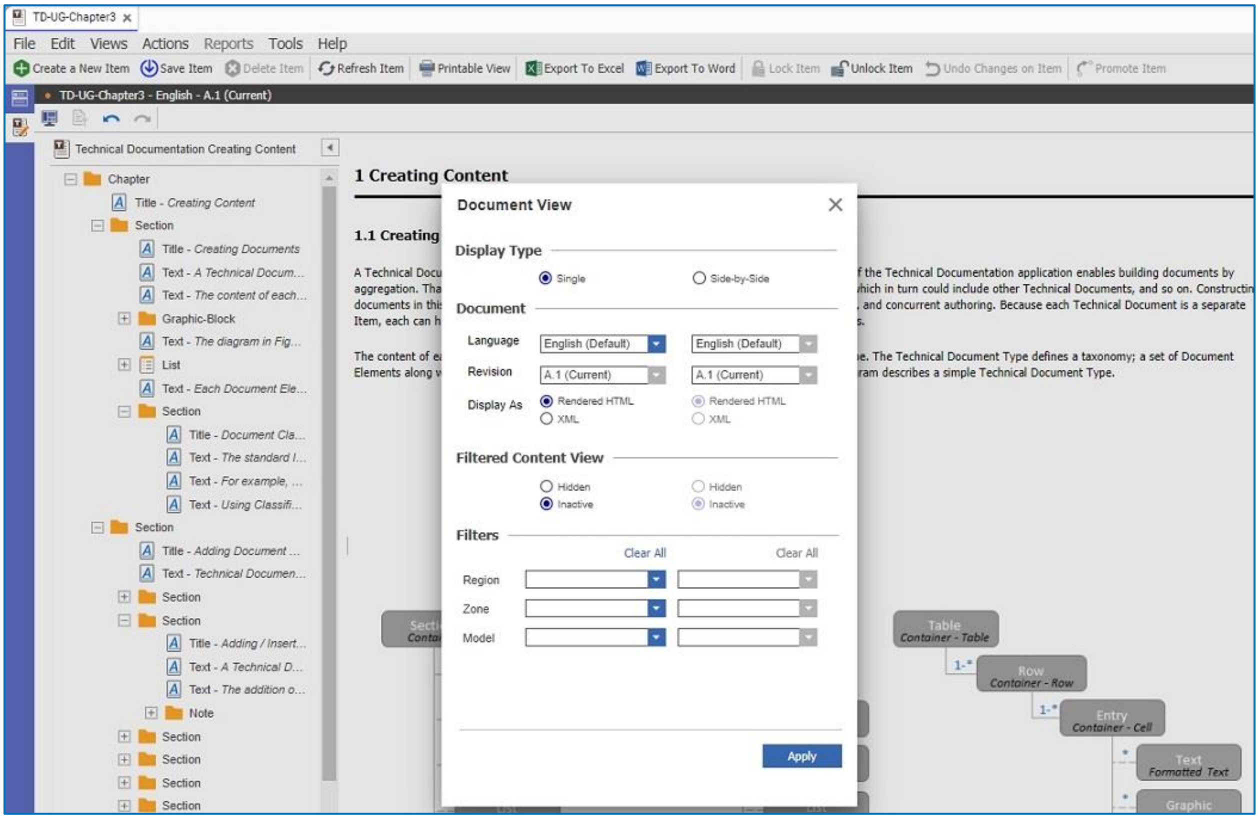



Figure 5.

2. Click the Side-by-Side button. The Language and Revision dropdowns are activated.

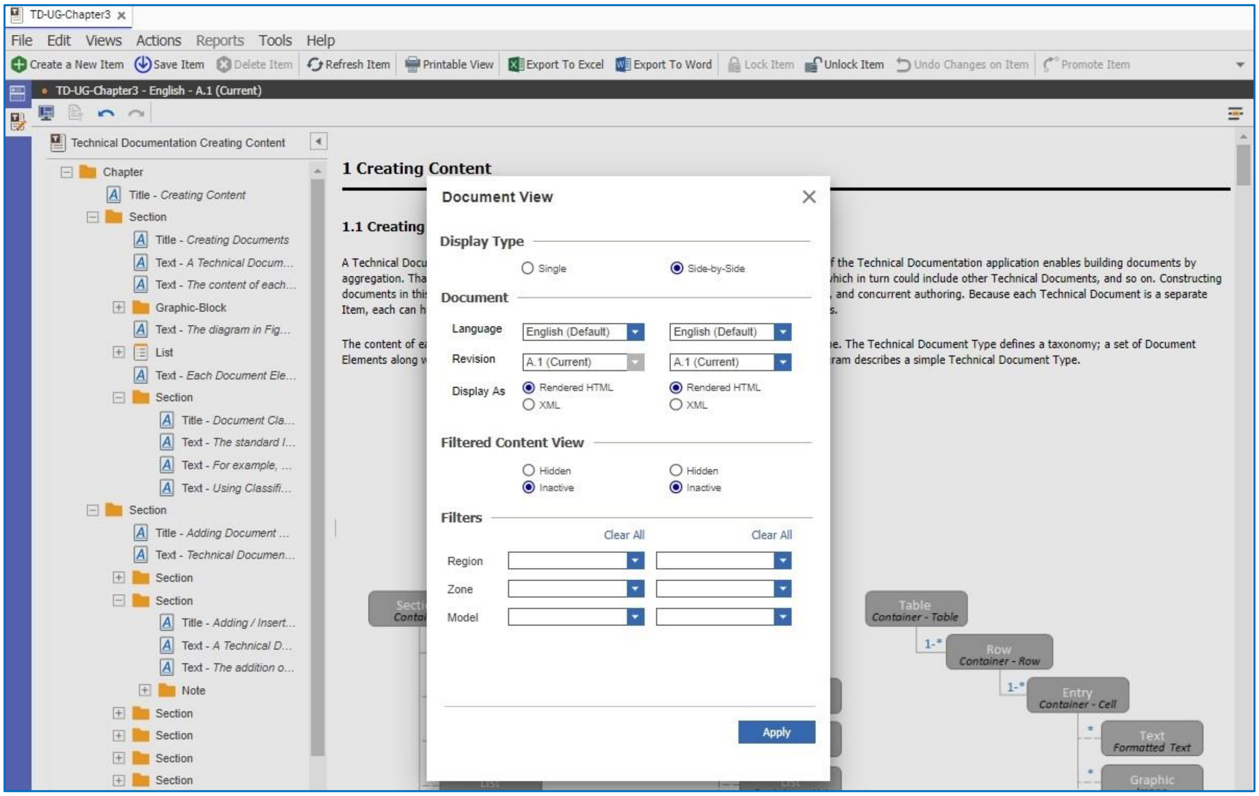



Figure 6.

3. Select the Language and the document version for both documents. You cannot specify the same document version and language for both documents. If you do an error message similar to the following appears:

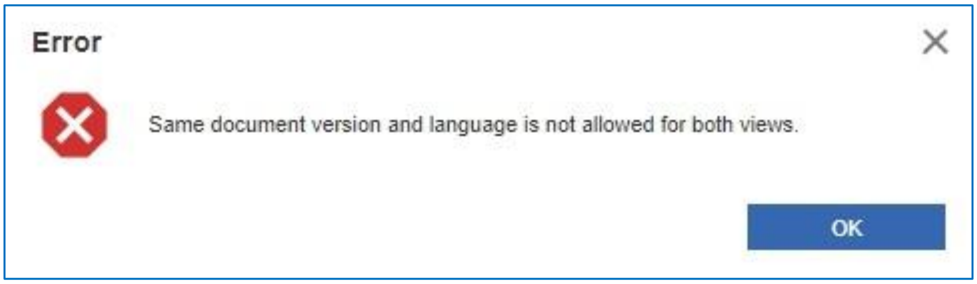



Figure 7.

4. Specify how the documents appear by selecting either **Rendered HTML** or **XML**. You can also select the filters to use for each document.

5. Press **Apply**. Both documents appear side-by-side.

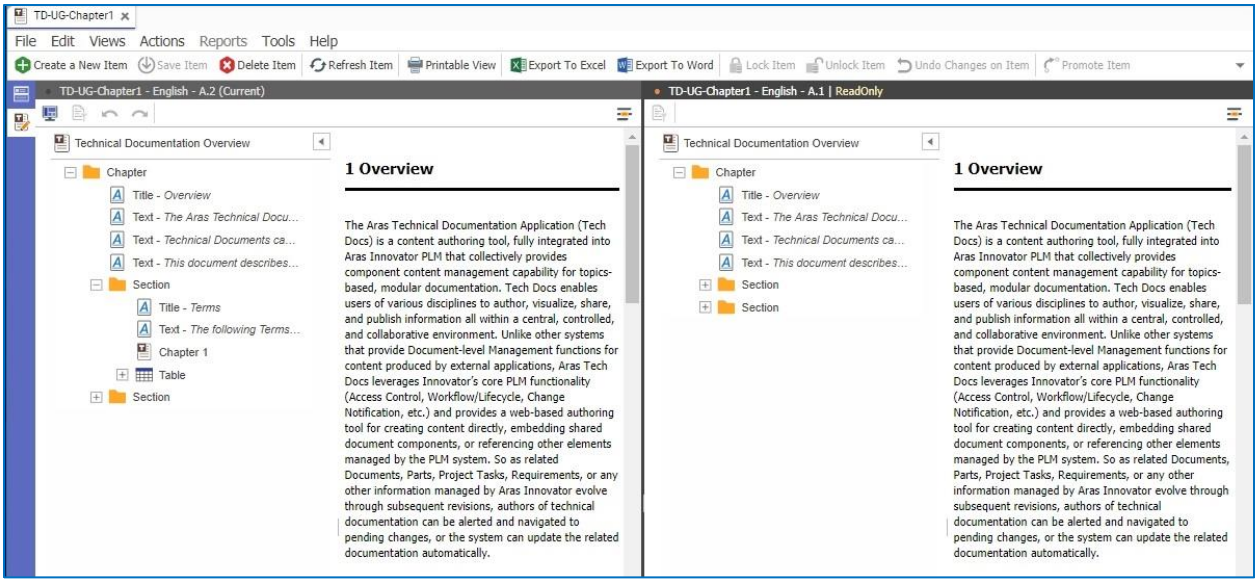


Figure 8.

The left side contains the document that you opened first in the main grid. The document that appears on the right is the one you selected in the Document View dialog box.

You can only edit the document that appears on the left side view which is marked as Current. The document on the right side is marked as ReadOnly. The context menus for each document are different. The context menu for the editable document displays the following items:

- Add
- Remove
- Copy
- Cut
- Convert to External
- Change Condition

The context menu for the read-only document displays the following items:

- Copy
- View Condition

## 2.4 Generating a Table of Contents

The Content Generator enables you to automatically generate a Table of Contents using the TOC-Section Document Element. The Content Generator for the TOC child Document Element iterates through a document and adds all the Chapter and Section document headings to the Table of Contents as shown here:


Figure 9.

**Note:** You can customize the number of headings displayed in the TOC by editing the TOC Item. For more information, refer to the *Aras Technical Documentation – Administrator's Guide*.

### 2.4.1 Adding a Table of Contents

Use the following procedure to add sections to the Table of Contents:

1. Right-click the Standard Doc folder in the Navigation pane. Select **Insert>TOC-Section** from the context menu.

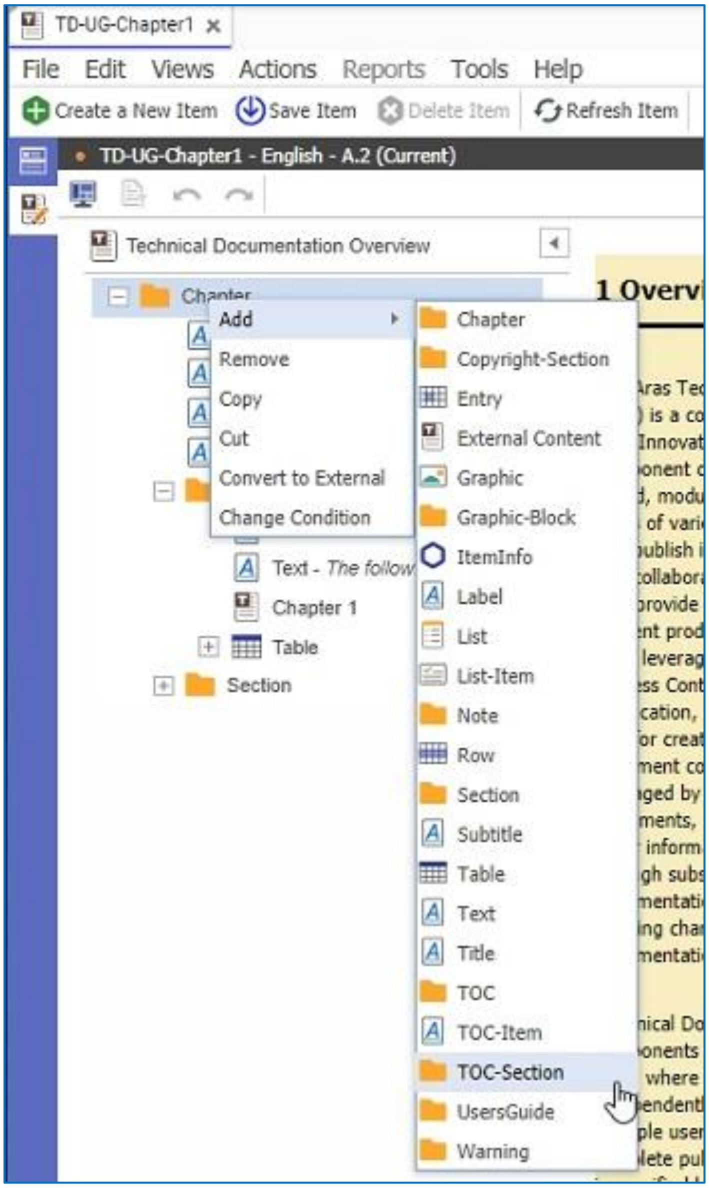



Figure 10.

A TOC-Section folder appears in the Navigation pane.

2. Select the Title Element and enter a title for the Table of Contents, as shown here:

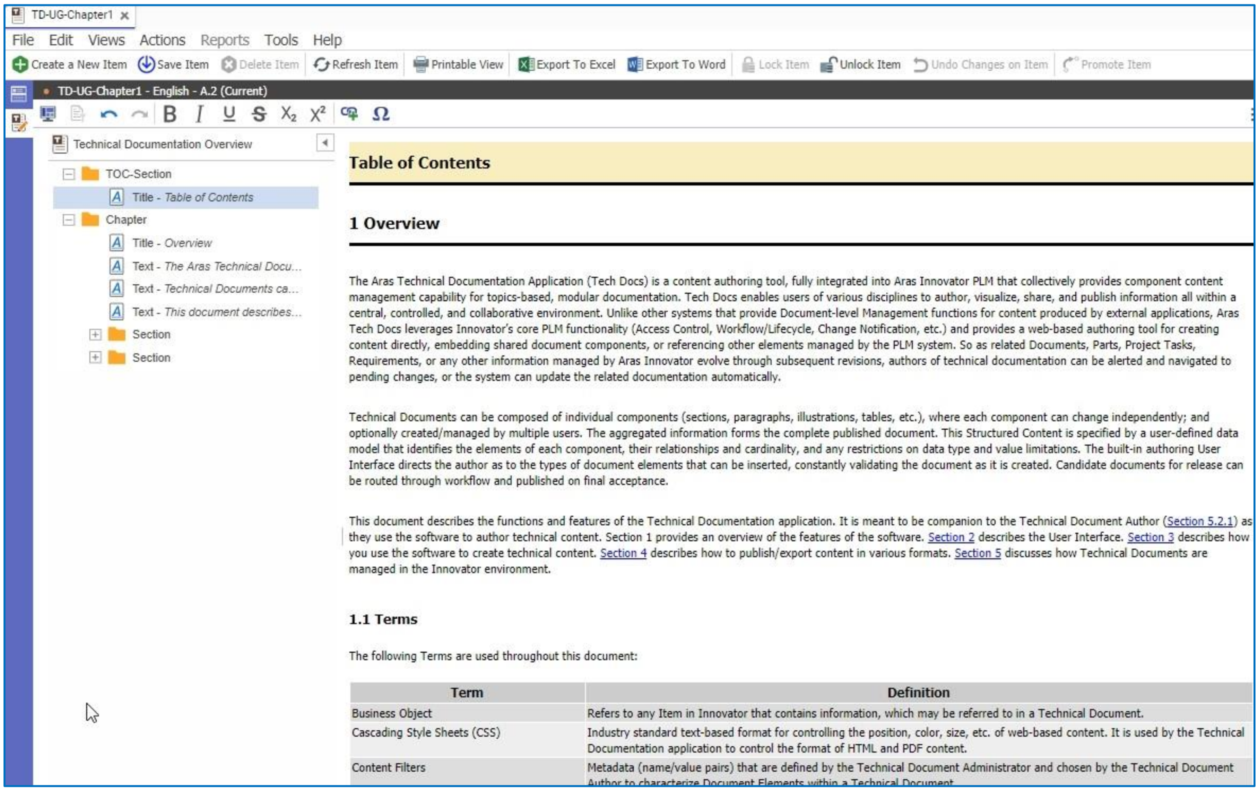



Figure 11.

3. Right-click the Title Element and select **Add>TOC** from the context menu to add entries to the Table of Contents.

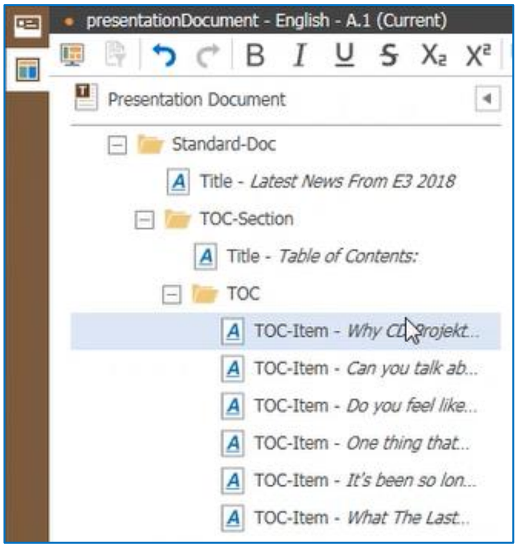



Figure 12.

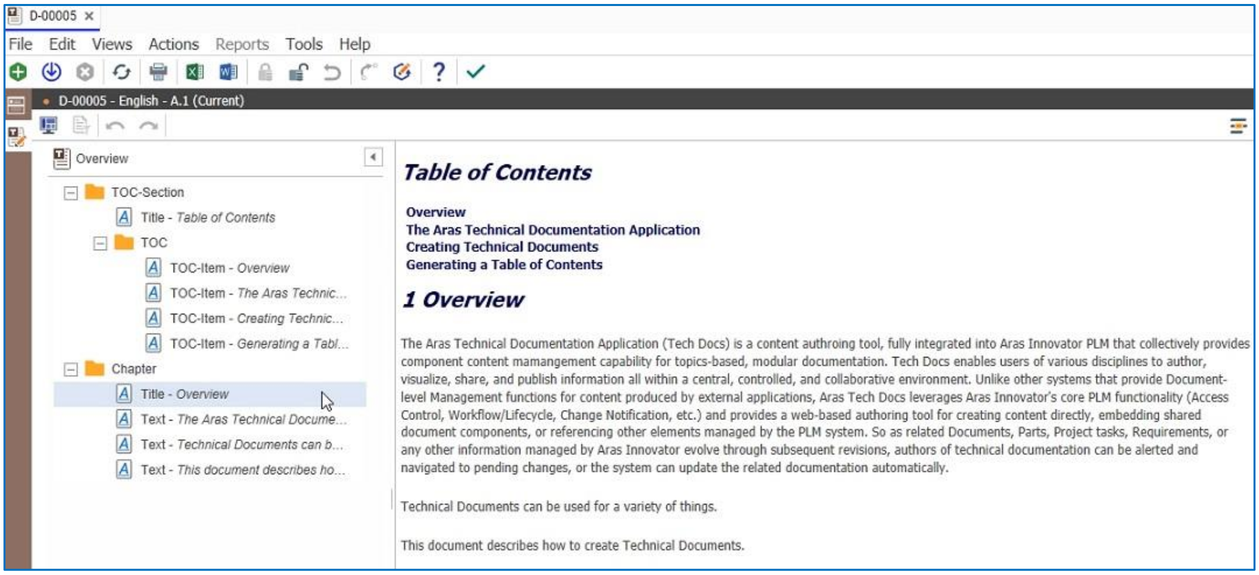

D-00005
File
Edit
Views
Actions
Reports
Tools
Help
D-00005 - English - A.1 (Current)
Overview
TOC-Section
Title - Table of Contents
TOC
TOC-Item - Overview
TOC-Item - The Aras Technic...
TOC-Item - Creating Technic...
TOC-Item - Generating a Tabl...
Chapter
Title - Overview
Text - The Aras Technical Docume...
Text - Technical Documents can b...
Text - This document describes ho...
Table of Contents
Overview
The Aras Technical Documentation Application
Creating Technical Documents
Generating a Table of Contents
1 Overview
The Aras Technical Documentation Application (Tech Docs) is a content authroing tool, fully integrated into Aras Innovator PLM that collectively provides component content mamangement capability for topics-based, modular documentation. Tech Docs enables users of various disciplines to author, visualize, share, and publish information all within a central, controlled, and collaborative environment. Unlike other systems that provide Document-level Management functions for content produced by external applications, Aras Tech Docs leverages Aras Innovator's core PLM functionality (Access Control, Workflow/Lifecycle, Change Notification, etc.) and provides a web-based authoring tool for creating content directly, embedding shared document components, or referencing other elements managed by the PLM system. So as related Documents, Parts, Project tasks, Requirements, or any other information managed by Aras Innovator evolve through subsequent revisions, authors of technical documentation can be alerted and navigated to pending changes, or the system can update the related documentation automatically.
Technical Documents can be used for a variety of things.
This document describes how to create Technical Documents.

Figure 13.

# 3 Creating Content

## 3.1 Creating Documents

A Technical Document is an instance of the Technical Document ItemType. However, the design of the Technical Documentation application enables building documents by aggregation. That is, one Technical Document can include one or more other Technical Documents, which in turn could include other Technical Documents, and so on. Constructing documents in this manner enables content reuse, allows for the separation of content by specialty, and concurrent authoring. Because each Technical Document is a separate Item, each can have their own access rights and evolve through independent workflows / lifecycles.

The content of each Technical Document is determined by the associated Technical Document Type. The Technical Document Type defines a taxonomy; a set of Document Elements along with the allowed hierarchy and cardinality of each. For example, the following diagram describes a simple Technical Document Type.

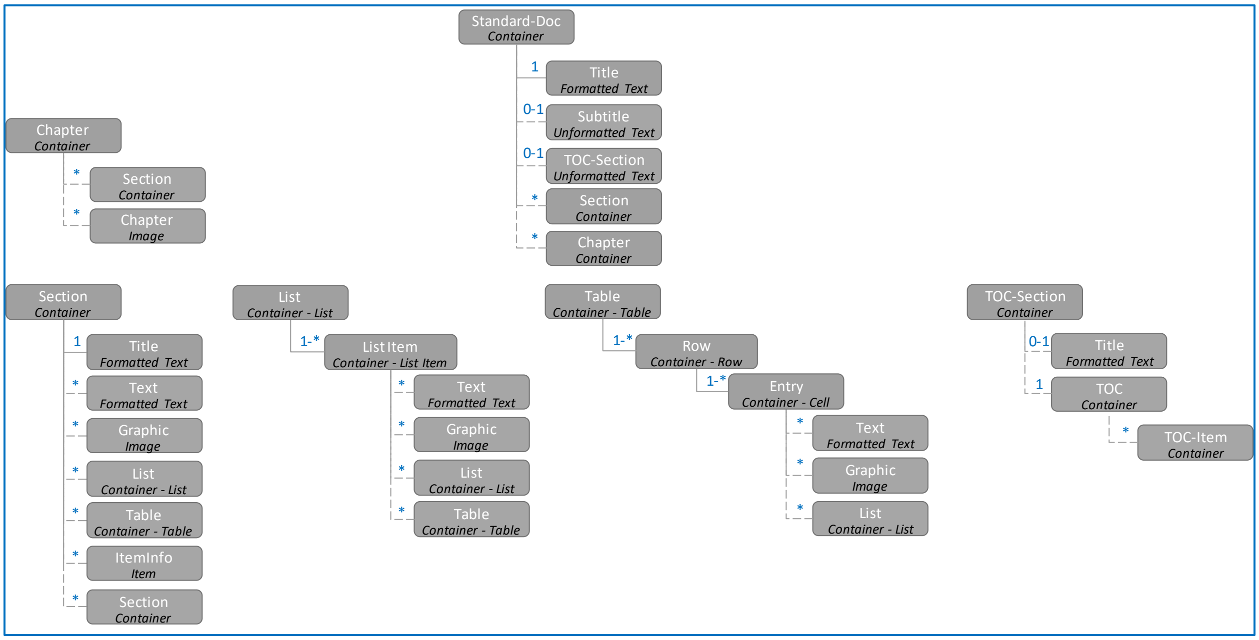



Figure 14.

The diagram in Figure 3 identifies the following Document Elements:

- Standard-Doc
- Title
- Subtitle
- Section
- Chapter
- TOC-Section

- TOC
- TOC-Item
- Text
- Graphic
- List
- List Item
- Table
- Row
- Entry
- ItemInfo

Each Document Element can either be of type Container, Formatted Text, Unformatted Text, Image, Item, Container – List, Container – List Item, Container – Table, Container – Row, and Container – Cell. These types are described in Section 3.2. Figure 3 also shows allowed hierarchies of Document Elements along with the number and sequence of Document Elements within them. For example, the Standard-Doc Document Element can contain the following Document Elements as direct children: a mandatory Title, an optional Subtitle, and any number (including 0) of Sections in this order. Likewise, if a Section Document Element is added, it must contain a Title followed optionally by either a Text, a Graphic, a List, a Table or an ItemInfo Document Element in any order and in any number (including 0 for each). These simple rules allow for elaborate and complex content structures.

### 3.1.1 Document Classification

The standard ItemType Classification property can be used to classify the content of a Technical Document. Each Classification value is assigned to the Technical Document ItemType. When used, a single Classification value can be bound to a single Document Element. When a Technical Document instance is created with a Classification value (see Figure 1 for a diagram of the Technical Document Form view) the Technical Document Element will ensure that the top-most node in the Document Element instance hierarchy for the document is set correctly.

For example, assume that an administrator added the Classification value of 'Document Section' to the Technical Document Element and associated that classification value with the Document Element 'Section'. When a Technical Document is created with the classification 'Document Section' the Editor will automatically insert a 'Section' Document Element as the root in the Document and disallow any peer Document Elements. That is, there will be one root Document Element – Section – only. All child Document Elements will be restricted based on what is defined by the Technical Document Type.

Using Classification in this manner will help in the organization of Technical Document components such that they can be searched more easily. Items with separate Classifications can also have alternate Forms associated with them. When an author would like to reuse a Technical Document, they will know that documents with the corresponding classification value will contain a single root node and thus can be inserted into an existing hierarchy of Document Elements wherever Document Elements of the type of the root node can be.

## 3.2 Adding Document Elements

Technical Documents consists of a collection of one or more Document Elements arranged in a hierarchy and sequence governed by the associated Technical Document Type. Each Document Element contains some type of content, e.g., text, images, or other Document Elements. Once the Document Element is created, the author can then proceed to add its content.

### 3.2.1 Context Menus

All Document Elements are created, removed, or updated through an explicit action executed by the selection of context-sensitive menus attached to each node in the navigation pane or set of content in the editor. Only changes that are allowed at each location will have corresponding menus included. For example, given the Document Structure defined in Figure 3, once a Title is added to a Document Element of type 'Standard-Doc' the author can only add a Subtitle or a Section. In this case the context menu for adding content to the document after the Title Document Element is placed will be at Subtitle or a Section.

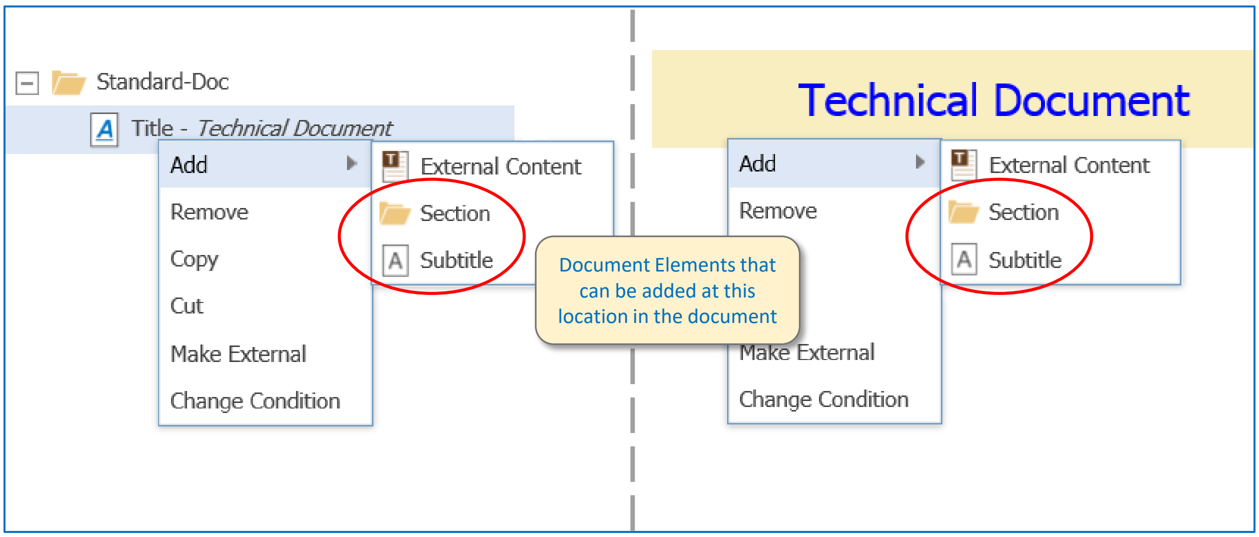



Figure 15.

#### *3.2.1.1 Handling the Enter Key*

When a Document Element is selected in the content editor/viewer, selecting the <Enter> key will display a context-sensitive context menu displaying the options associated with adding content at that location. For example, instead of using the right-mouse button display the context menus in Figure 4 if the author selects the **Enter** key the following context menu will be displayed:

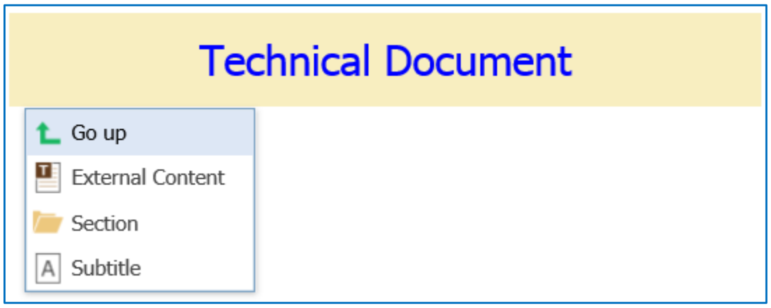



Figure 16.

The menu options displayed are for Adding content. Recall that the term 'Add' refers to adding peer Document Elements. The top menu item ('Go up') is used to display the 'Add' menu items for the parent Document Element. This functionality is most useful when adding content within a List.

### 3.2.2 Adding/Inserting

A Technical Document consists of a hierarchy of Document Elements that collectively form a lineage of Document Elements with potentially multiple branches of arbitrary depth. There can be one or more root (or top-most) elements each containing child elements, which can in turn contain child elements and so on. The complete document element tree is exposed in the navigation pane in the editor.

The addition of Document Elements to a Technical Document is usually done in the context of another, existing Document Element. A new Document Element is either a peer or a child of the existing element. In the nomenclature of the Technical Documentation application 'Adding' refers to the addition of a peer and 'Inserting' refers to the addition of a child. For each existing Document Element in a document if it can have peer elements an **Add** sub-menu will exist in the context menu (see Figure 4) with sub-menus items for each allowed Document Element. Likewise, if an existing Document Element can have child elements, an 'Insert' sub-menu will exist in the context menu.

**Note:** Menu option 'External Content' refers to selecting an existing Technical Document instance, the content of which will be placed as either a peer or child to the selected Document Element depending on the menu selection.

### 3.2.3 Text

All text-based content for a Technical Document is included in one or more Document Elements. These Document Elements are configured to either include formatted or unformatted text. Note that most of the style and positioning rules for text-based content should be defined for the Document Element as part of the Technical Document Type Style. In this case, authors simply add the text, and the system applies the appropriate style (color, size, position, margin, etc.) automatically.

Once any Document Element that is of type formatted text or unformatted text is added to a document, a text entry area is added to the Content Editor with the default 'Text…' displayed as a placeholder. Once the author begins typing text characters, they are displayed in succession in the entry area for the element. As text is typed the text for the Document Element node in the navigation pane is updated. Any text that is too long to display in the navigation pane is truncated with an added ellipsis. When typing Text content, selecting the Enter key will result in the automatic creation of a Text Document Element and subsequent characters will be added to the new Document Element accordingly.

#### *3.2.3.1 Formatted Text*

Formatted text can have a limited set of styling applied by the author. In general, style for text-based content should be specified by the style associated with the Technical Document Type. When any formatted text-based document is selected the toolbar for the Editor is updated to display additional buttons that can be used to apply the following style to the currently selected text (if any text is highlighted) or will apply to newly typed characters starting at the location of the cursor in the text area:

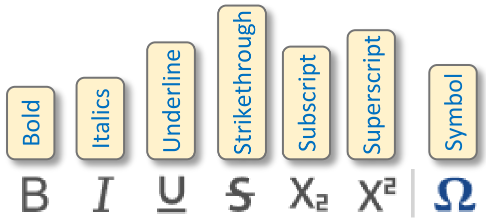



Figure 17.

Other than symbols, the formatting options are toggles; they either are applied to newly typed text or not. Each of these can be applied collectively. For example, text can be both bold and italics. Un-applying any of these formats is simply a matter of selecting the text and selecting the appropriate format button to undo the application of the format.

#### *3.2.3.2 Unformatted Text*

Unformatted Text cannot have any additional style applied by the author. It can, however, have style automatically applied as defined by the style associated with the Document Type.

#### *3.2.3.3 Added Character Symbols*

A set of additional character symbols is added to the editor and accessed by selecting the Symbol button in the editor (Figure 5) and selecting the character button in the pop-up tooltip. These additional characters can only be applied individually to formatted text Document Elements.

#### 3.2.3.4 Automated Text Document Element Placement

In any given Document Element context, if there exists a child Document Element of type formatted or unformatted text (as defined by the Technical Document Type definition) it will be automatically inserted when the author begins to type in the editor. For example, using the document structure as defined by Figure 3, when the author places a *Standard-Doc* Document Element in the document and subsequently begins typing text, a *Title* Document Element will automatically be added, and the typed text automatically added as content. In this case, the system 'knows' which document element to automatically insert because the rules of the Technical Document Type describe it - a *Title* Document Element must exist as the first child element under the *Standard-Doc* element. Because this element is a text-based element, the typed text is added to the *Title* document element after it is added by the system. As a second example, a *List* Document Element contains one or more *List Item* Document Elements. After a *List Item* is added and the author begins to type text, a *Text* Document Element is automatically inserted, and the typed text added as content. In this case, the *Text* Document element is the only text-based document element for the *List Item*. The system assumes that when the author begins typing text that a *Text* Document Element should be placed.

To a certain extent, this automated behavior helps to streamline the authoring process and attempts to emulate (as much as possible) common word-processing behavior. Since the Technical Documentation application is a structured document application, all content must be attributed with specific Technical Document Types. Thus, any text must be included within a Document Element.

#### 3.2.3.5 Embedding Newlines/Carriage Returns

Text-based content does not include newlines/line breaks by default. This assumes that any paragraph will exist within its own Document Element. The default handling of the **Enter** key is to display a context menu (Section 3.2.1.1) or to automatically place a new Document of the same type when such a Document Element is allowed by the associated schema. The method for embedding a line break within a text-based Document Element is to use the **Shift-Enter** key combination. Doing so will result in a newline/carriage return added at the current location of the cursor.

#### 3.2.3.6 Copying and Pasting Text

Text content can be cut/copied from one Document Element and pasted within another. In this case, all formatting will be retained in the pasted text. In addition, text can be copied from another application and pasted within a Document Element. Text content copied from an external application will not retain its formatting. Any copied text content is pasted within the system's copy buffer and can thus be used as a transfer mechanism to and from a Technical Document.

To copy text, select and drag the cursor using the left mouse button over the text to be copied in the content Editor/Viewer. The selected text will be highlighted. The Ctrl-C key combination will copy the highlighted text to the copy buffer. Note the **Copy** menu item in the document element's context menu is used for copying the Document Element, not the selected text.

To place copied text, place the cursor within the text area of an existing Document Element and use the **Ctrl-V** key combination. The text in the copy buffer will be inserted at the cursor location. Note the **Paste** menu item in the document element's context menu is used for pasting a copied Document Element, not the text in the copy buffer.

### 3.2.4 Lists

Lists within the Technical Documentation application are implemented using Document Elements based on a List Type. Lists are essentially containers with an unlimited number of *List Item* Document Elements.

When a *List* Document Element is created its *List Items* can be rendered as either bulleted, numeric, or alpha. The default is bulleted. Selecting either a *List* or *List Item* Document Element will expose the following toolbar menu buttons. The selected option will have a box around it. All *List* Document Elements displayed in the navigation pane will be displayed with the List icon.

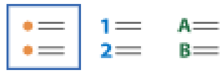


Figure 18.

#### *3.2.4.1 List Items*

*List* Document Elements are expected to have child *List Item* Document Elements as the only element content. *List Item* Document Elements act as the container for all content that can be included. For example, the document structure defined in Figure 3 identifies the Document Elements that can be included in a *List Item*: *Text*, *Graphic*, other *List*s, and *Table*s in any order and in any number. All *List Item* Document Elements displayed in the navigation pane will be displayed with the List item icon.

### 3.2.5 Tables

Tables within the Technical Documentation application are implemented using Document Elements based on a *Table* Type. Tables are essentially containers with an unlimited number of *Row* Document Elements.

Tables in Technical Documents are restricted to be 'rectangular'. That is, every row must contain the same number of cells/columns and every column must contain the same number of rows. This restriction is supported in the table-specific context sub-menus that are displayed in the content editor. For example, a *Cell* Document Element cannot be added to one *Row* without a new Cell be added to all Rows. As a result, only Row and Column operations are displayed in the context menus.

When a Table Document Element is added the author is presented with a dialog to choose the number of rows and columns.


Figure 19.

The default number of rows and columns is 2, with a maximum of 10 rows and columns for each initial value.

**Note:** The Technical Documentation application is not designed to create tables with large numbers of rows and/or columns. For large tables, it's better to use a Content Generator for automatically creating each row and cell and add content based on user-defined logic.

Tables are rendered in the content viewer/editor based on the Style settings for the associated Technical Document Type. The out-of-the-box Style configuration uses a single black border with alternating background colors for the cells in even/odd rows.

#### *3.2.5.1 Adding/Removing Columns and Rows*

*Table* Document Elements have specific context menus for adding and removing columns and rows. The Table context menu will only be displayed when either a *Table*, *Row*, or *Cell* Document Element is selected.

**Note:** When selecting content within a table, note that the Document Element selected depends on where in the content viewer/editor the mouse cursor is placed.

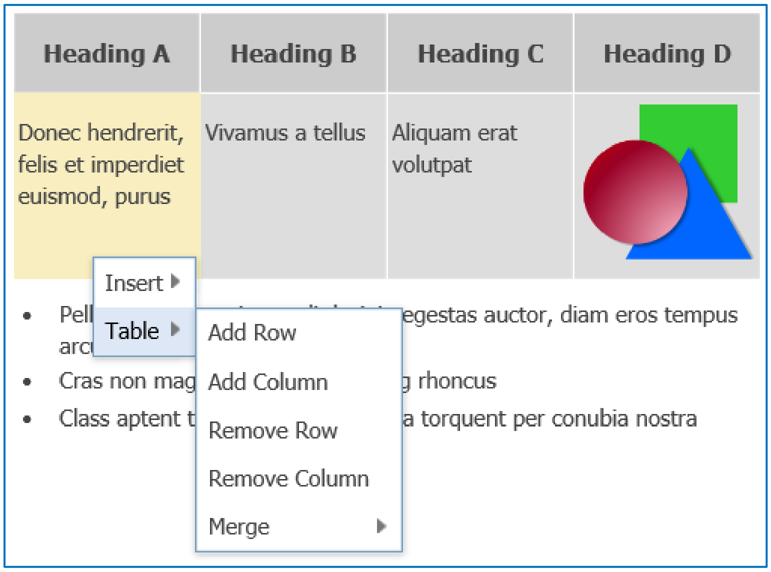



Figure 20.

When adding a row or a column to a table a row/column is always added below or to the right respectively. Note that the Table context menu will be different depending on which type of Document Element is selected. In Figure 9, a *Cell* Document Element was selected in the content editor. In this case, the **Insert** context sub-menu will display a list of Document Elements that can be inserted into the cell. The Table sub-menu (as shown) displays menu items for modifying the table at that selected location in the table. For example, adding a column would add a new column between the columns with header titles 'Heading A' and 'Heading B'. Adding a row would add a new row to the bottom of the table.

#### *3.2.5.2 Merging/unmerging cells*

The Content Editor supports the ability to merge and unmerge cells in a table. Merge/Unmerge functions are applied one cell at a time. That is, to create one cell across all columns in one row, the author must first merge two adjoining cells and then merge that with neighboring cells and so on. Also, merging cells that are already merged with a neighboring row or column requires the adjoining cells to have been merged prior. Using Figure 10 as an example, to achieve the table configuration in the last step the first two cells in the second row are merged, followed by the first two cells in the third row. At this point, the first (merged) cell in the second row can be merged with the first (merged) cell in the third row.

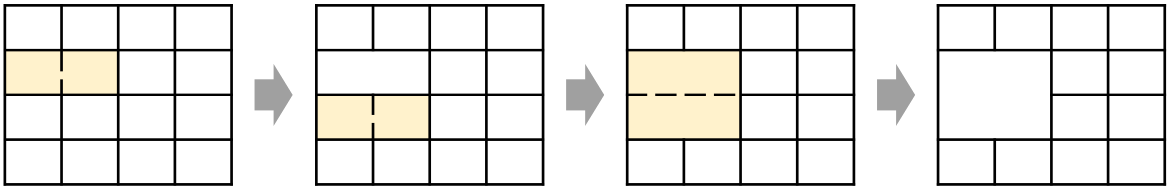


Figure 21.

The **Unmerge** function will unmerge all cells that are merged for the selected cell. That is, to apply an Unmerge function to the merged cells in the last table in Figure 10 would return the table to the original state as shown in the first table.

### 3.2.6 Links

Text content within a Technical Document can be linked to any Document Element within the same Technical Document or within a separate Document Element. In addition, text content can also be associated with a URL, like a hyperlink in an HTML document. Both types of links behave similarly when the content with these links are exported/published (see Section 4). Once the link is created, the Content Editor can be used to explore the linked content. Links between Document Elements establishes a Relationship between the two corresponding Technical Document Items. Thus, linked Document Items cannot be removed if they are the target of an existing link.

### 3.2.7 Creating a Link to a Document Element

To create a link to a Document Element:

1. Select the text from within an open Technical Document in Edit mode.
2. Place the cursor over the beginning of the text and drag with the left mouse button pressed to dynamically select all text characters/words associated with the link.
3. Select the **Create Link** button in the toolbar.

   The Link Editor Dialog will be open and display the contents of the currently opened Technical Document at the top.

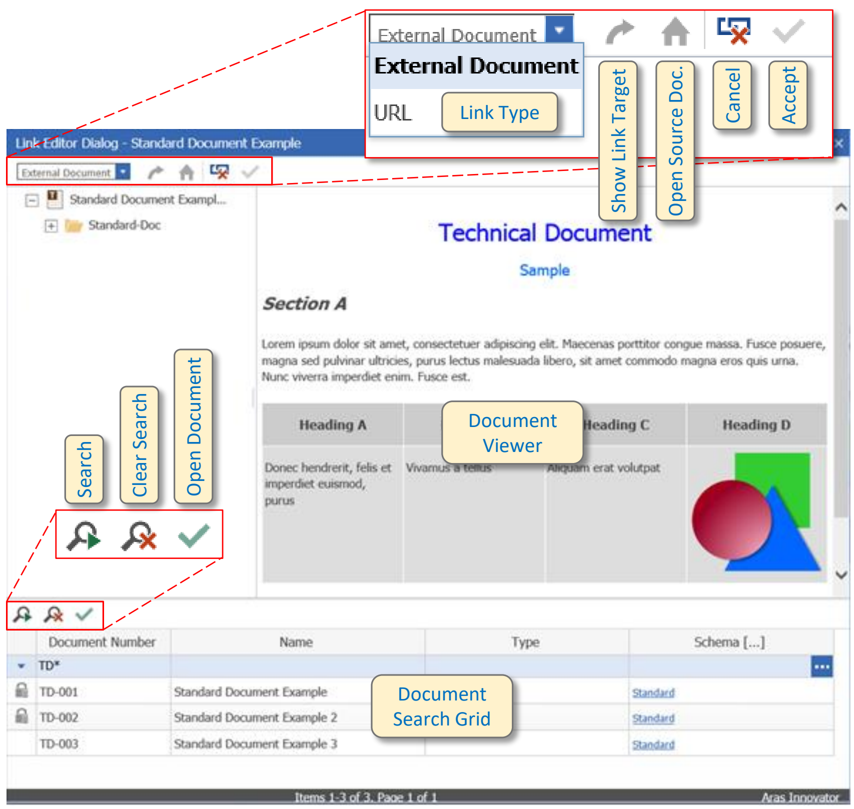



Figure 22.

4. Select the Document Element from the current document to set as the link target or search for another document using the search grid at the lower portion of the Link Editor Dialog

   Document Elements are selected the same as in the Editor; either by selecting a node in the Navigation Pane or selecting the element in the content viewer. If another document is searched, select the **Open Document** button (green check mark) above the search grid to display the selected document.

5. Select the **Accept** button (green check mark) at the top of the dialog to accept the link target

   Doing so, will close the Link Editor Dialog. The text used for the link in the source document will be highlighted to indicate that the text is associated with a link.

| | |
|---|---|
| **Note:** | In the out-of-the-box version of the Technical Documentation application, text with a link associated with it is rendered in blue with an underline. |

#### *3.2.7.1 Creating a Link to a URL*

1. To create a link to a URL:

   Select the text from within an open Technical Document in Edit mode.

2. Place the cursor over the beginning of the text and drag with the left mouse button pressed to dynamically select all text characters/words associated with the link.
3. Select the **Create Link** button in the toolbar
4. Select 'URL' from the Link Type combo box in the top portion of the Link Editor (Figure 11)
5. Type the URL for the link in the text edit box.
6. (Optional) Select the **Check URL** button to open a separate browser window using the URL that was typed

#### *3.2.7.2 Modifying and Testing a Link*

Once a link has been created, the target content can be modified by selecting the **Edit Link** button on the toolbar. This button, along with the **Test Link** and **Remove Link** buttons, will be displayed when text containing a link is selected in the content editor.

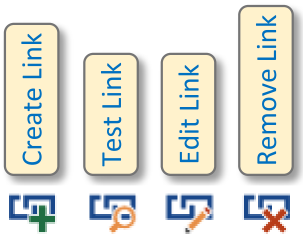



Figure 23.

When the **Edit Link** button is selected, the Link Edit Dialog is displayed. At this point, the author can select the Document Element (or URL) to use like the steps in Section 3.2.7.1 and Section 3.2.7.2.

To test a link, select text containing a link and select the **Test Link** Button. The Link Edit Dialog is displayed with the Technical Document (containing the target Document Element) shown and the Document Element target selected. If a URL was used for the target, a browser will be opened to the selected URL.

### 3.2.8 Item Document Elements

*Item* Document Elements are Document Elements with a Relationship to an Item within Innovator. A Relationship (of type tp_ItemReference) is created between the Technical Document and the selected Item. A referenced Item can be either a Part, Document, or CAD Document.

**Note:** Administrators may choose additional items that a Technical Document can display. For more information, refer to Aras Technical Documentation – Administrator's Guide.

When an Item Document Element is added to a Technical Document, the author is prompted to select an existing Item. Adding Document Elements of this type to a Technical Document enables the author to establish a direct relationship to Items that, presumably, the document is written for. Once included in a Document, the author can view the associated Item by selecting the node of the *Item* Document Element instance in the Navigation Pane and select the 'View Item' sub-menu. Doing so will open the main Form associated with the Item. Nodes of this type will be displayed in the Navigation Pane using the icon associated with the related Item.

*Item* Document Elements typically have Content Generators associated with them (Section 3.6). In this case, the content created by the Content Generator is added to the document automatically. *Item* Document Element instances together with Content Generators allow authors to quickly and efficiently created document content dynamically and enable the synchronization between the two.

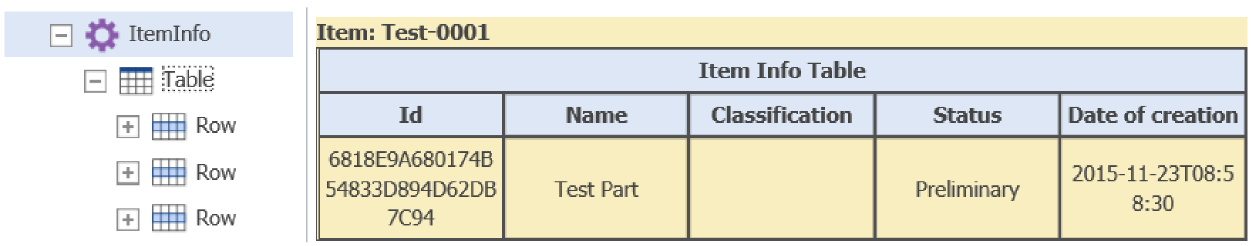



Figure 24.

#### *3.2.8.1 Mapped Properties*

Mapped Property Document Elements are a type of Unformatted Text Document Element that are used to reference a Property of an Item referenced by an ancestor Item Document Element. When authors place instances of a Mapped Property Document Element, the system will automatically apply the content of the associated Property of the referenced Item.

Mapped Properties are configured with Document as part of the associated Document Type. Mapped Property Document Element have two Attributes: 'Property' and 'Mode'. The **Property** attribute specifies the name of the Property that is mapped. The **Mode** attribute specifies whether the Property (and thus the related Item) can be modified within the Technical Document.


*Figure 1: Attributes Dialog for Mapped Property Document Elements*

Mapped Property Document Elements are edited using a pop-up Form in the Technical Document Editor. To edit a value, select the Document Element in the Editor or Tree View and/or double-click to display the Property Editor Form. Document Elements with a grey, dotted border can be edited. Document Elements with a red, solid border are read-only. For example:

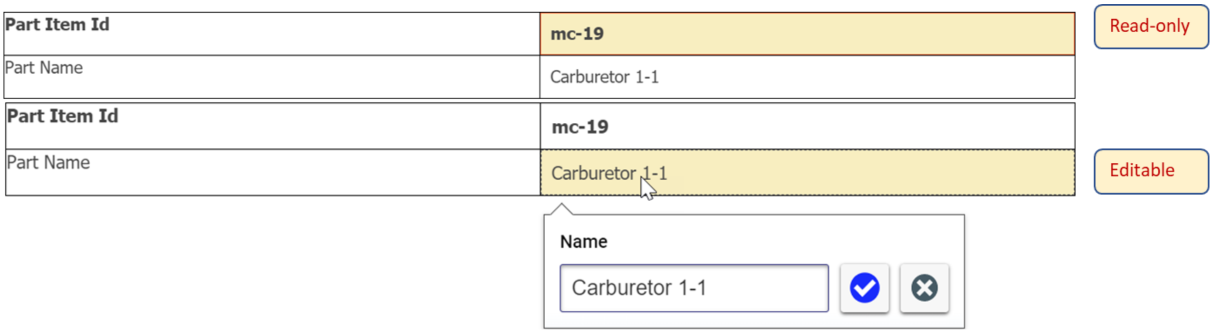



*Figure 2: Mapped Property Editor Form*

Values can only be changed using this Form. Selecting the '**Check**' button will update the Mapped Property Document Element in the technical Document only. Authors will see a pencil icon (), signifying that the Property of the referenced Item has not been updated, next to the Document Element in the Tree View. The referenced Item will be updated when the technical document is saved, in which case the pencil icon will be hidden. Selecting the '**X**' button will cancel the update without any changes to the technical document.

**Note:** Selecting the pencil icon () in the Tree View will prompt the author with a Dialog to discard the updated Property. In this case selecting 'OK' will revert the modified Property(ies) to their previous value.

The following Property Value Types are supported:

- String
- Text
- Numeric (Integer, Decimal, etc.)
- Date
- Boolean
- Item (Display only - Editor not available)
- List (Display only - Editor not available)
- Image (Display only - Editor not available)

The Editor can display multiple Property Types, as shown above, but not all can be edited with the Technical Document Editor. Authors will be presented with an error if an attempt is made to edit an unsupported Type:

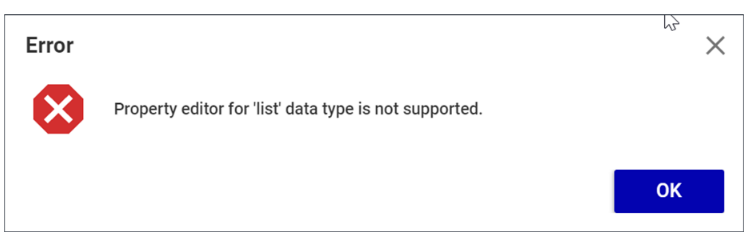



*Figure 3: Mapped Property Editor Error Dialog*

The Edit Field in the Property Edit Form is based on the Property Type. For Text and Numeric Fields, an attempt to set a value that does not match the type will result in an Error being displayed in a Dialog.

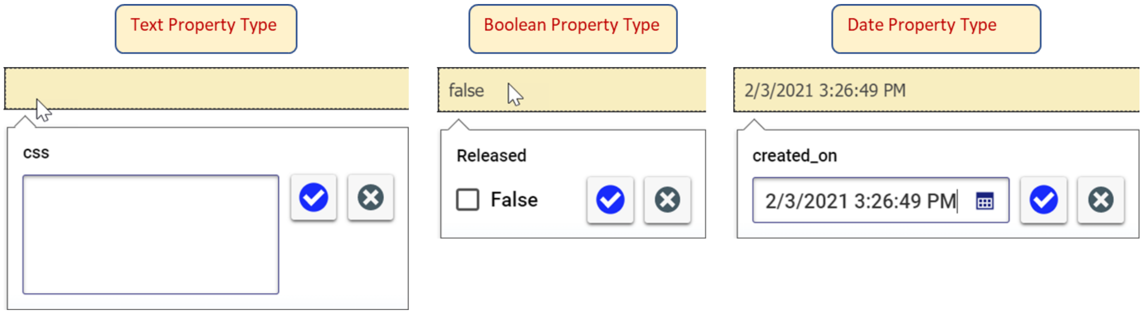



*Figure 4: Mapped Property Edit Forms*

Modifying a versionable Item vis-à-vis Mapped Property Document Elements will result in the referenced Item to be versioned automatically. Since Technical Documents refer to Items via fixed Relationships, the Item Document Element will show as stale since it will refer to the previous generation of the Item.

## 3.2.9 Graphics

*Image* Document Elements refer to any 2D raster or vector-based image that is supported by Innovator. All graphical content for a Technical Document must be stored as a separate Graphic Item. The Technical Document Application includes a 'Graphics' TOC folder that includes Graphic instances. Graphics have the following Form:

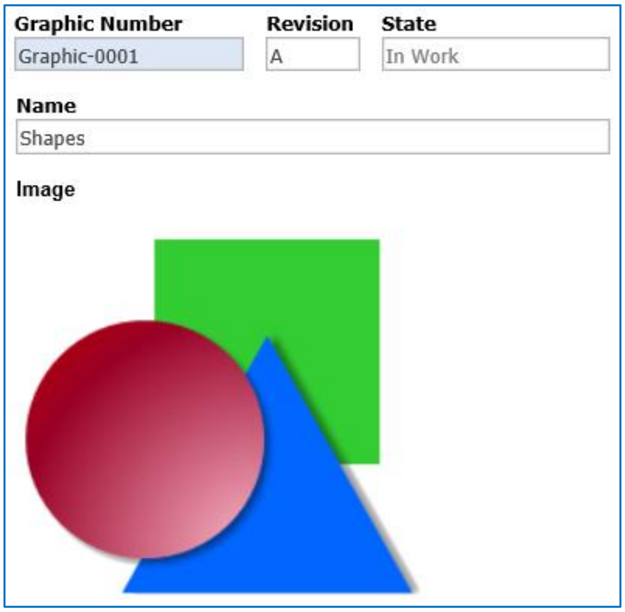



Figure 25.

### *3.2.9.1 Adding an Image*

To add an image to a Technical Document, Add or Insert any *Image* Document Element Type. The user will be presented with a Search Grid for Graphics. Selecting an instance will add the corresponding image content at the selected location with the document. In addition, a Relationship is added between the Technical Document and the selected Graphic.

## 3.3 Filters

Filters provide a mechanism for Technical Document Authors to associate Characteristics (metadata) with Document Elements and use these characteristics to automatically filter the content. This is useful when technical documents can serve multiple, but related, purposes. For example, a technical document that describes several similar models of the same part. Content that is specific to any model (or models) can have filters associated with it. When an end user wishes to see only the content that is either associated with all models or is particular to a specific model, they can identify the appropriate filters and the Technical Document Editor will display the correct content accordingly.

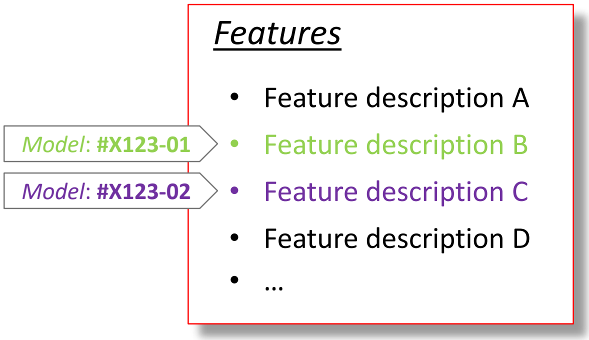



Figure 26.

Referring to Figure 26, the list of features associated with Model 'X123-0***1***' is A, B, and D. The list of features associated with Model 'X123-0***2***' is A, C, and D.

The list of available *Filter Names* and associated *Filter Values* is set by the Technical Document Administrator and exists for all Technical Document Types. By default, the following Filters are included with the Technical Documentation Application: Region, Zone, and Model.

### 3.3.1 Adding a Filter to a Document Element

To add a Filter to a Document Element, right-click the Document Element either in the Navigation Pane or in the Content Viewer and select Change Condition. The Change Element Condition dialog is displayed.

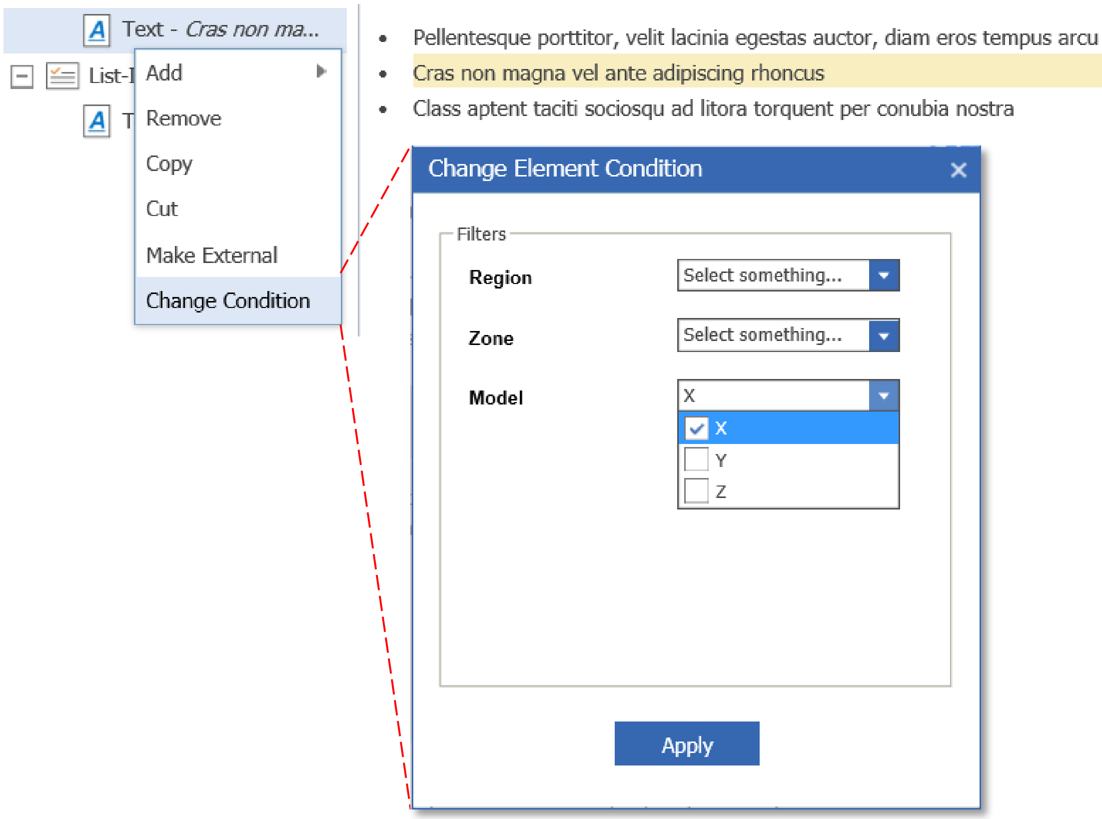



Figure 27.

The Change Element Condition dialog will display each Filter that has been configured for the Technical Documentation Application. Each Filter Name is displayed with a combo-box list containing each of the corresponding Filter Values. In Figure 27, the 'Model' Filter has the following Filter Values: 'X', 'Y', and 'Z' with the Filter Value 'X' selected. If this were the only Filter selection, then the selected Text Document Element would have the Filter 'Model: X' associated with it.

**Note:** A Document Element can have multiple Filters applied and/or multiple Filter Values for each Filter Name. Selecting the checkbox next to the Filter Value selects and adds the Filter Name/Value to the selected Document Element.

A set of Filters can be applied to the entire Technical Document. To do this, right-click the root node in the Navigation Pane (it will be displayed at the top with the name of the document) and select **Change Condition** in the context menu.

Document Elements with associated Filters associated with them are displayed in the Navigation Pane with a Filter icon added to the right of the node. This alerts the author that the corresponding content could be disabled or hidden with the application of a Filtered View.


Figure 28.

### 3.3.2 Creating a Filtered View

A Filtered View contains content that is either hidden or disabled based on the application of one or more Filters. The set of Filters is applied to all content that is displayed in a Technical Document, including referenced content. Only Document Elements that have a Filter applied will be affected by a Filtered View. This is because the system assumes that content without Filters apply to the set of all Filters.

To create a Filtered View, select the **Document View** button on the toolbar. Doing so will display the Document View dialog. The Document View dialog provides a list of all Filters showing each Filter Name and corresponding Filter Values in a combo-box list. To apply a Filter to the view, select the check mark next to the Filter Name. This will enable the list of Filter Values.

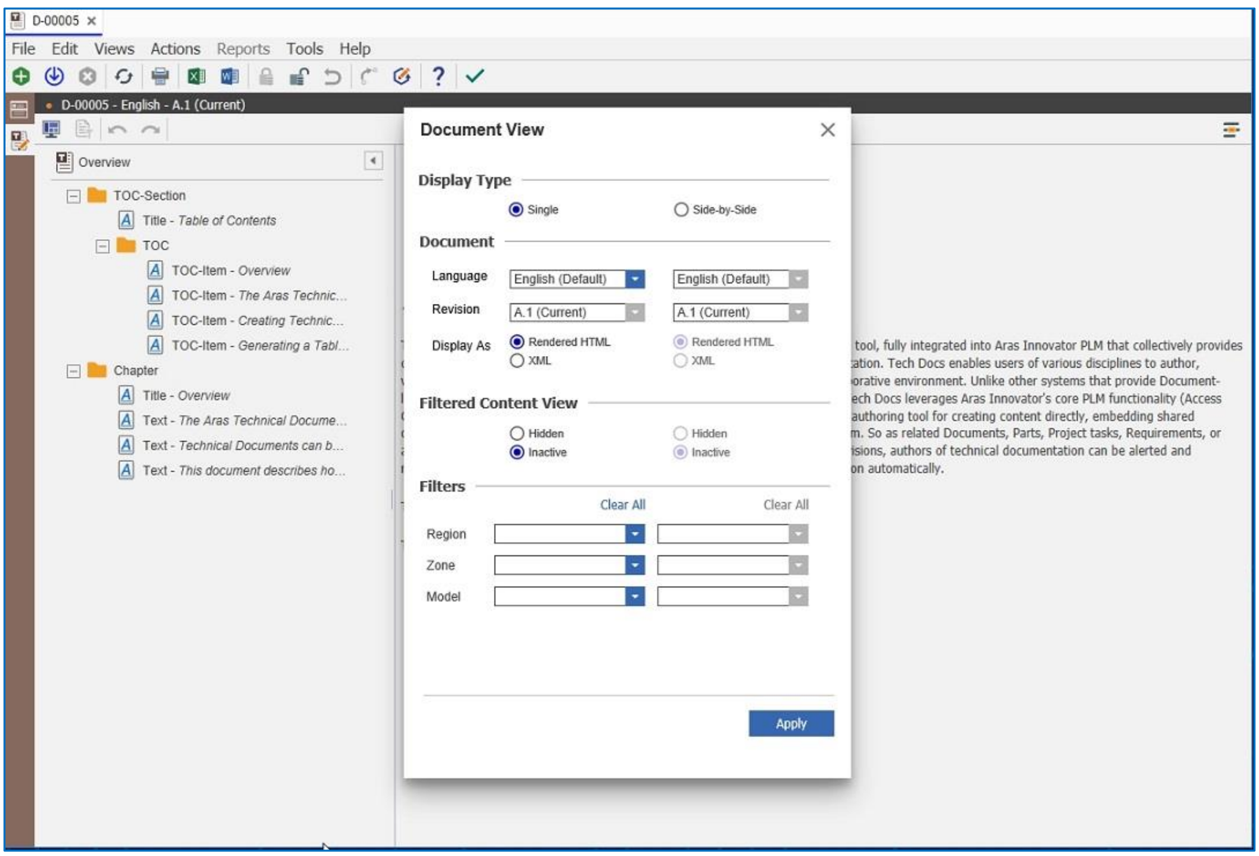



Figure 29.

The Filter Values that have been assigned to Document Elements in the current Technical Document will be highlighted (in color) in each list. To apply a Filter Value, select one or more Filter Values in each list and select the **Apply** button. By default, content that doesn't apply to the selected list of filters is displayed as disabled. Document Elements that have been disabled in a Filtered View will be displayed lighter (partial transparency) in the Navigation Pane and as a collapsed section in the content viewer with the Filtered Icon displayed to the right (Figure 30). To expand the disabled section, select the '+' button next to the disabled content in the content view.

| | |
|---|---|
| **Note:** | Selecting the Filter Icon to the right of disabled content in the Content Viewer will display the Change Element Condition dialog for the Document Element. In this case, the author can change the associated Filters for the disabled Document Element. |


Figure 30.

To hide disabled content in a Filtered View, select the radio button next to the 'Hidden' label in the Document View (Figure 29). In this case, disabled content will not be shown in the Content Viewer.

The Document View dialog allows the author to select a set of Filters for a Filtered View. To turn on/off a Filtered View, select the **Enable Filter** button in the toolbar. Note that this button is selected by default when the user selects the **Accept** button in the Document View dialog.

## 3.4 Groups

Grouping Document Elements allows for a set of Filters (Section3.4) to be applied to all content within the Group. Grouped content is displayed in the Navigation Pane using the Group Icon with all content within the group displayed as direct child nodes. There is no visual indication in the Content Viewer for grouped content. Any contiguous set of Document Elements can be grouped that exist within a Document.

### 3.4.1 Grouping/Ungrouping Document Elements

To Group Document Elements, select each in the Navigation Pane using the left mouse button while holding down the **Control** Key. Note that only contiguous Document Elements can be grouped. When all Document Elements are selected, right-click on one of the selected Document Elements to display the context menu and select **Group**. Any grouped content can be ungrouped by right clicking the *Group* Document Element and selecting **Ungroup** in the context menu.

## 3.5 Creating and Referencing External Content

Technical Document content can be an aggregation of multiple Technical Documents, which in turn can contain other Technical Documents and so on. The result is a hierarchy of document components. Individual Technical Document components can be reused in other Technical Documents and be edited concurrently with the documents that refer to them. A Relationship instance is created between the source document (document which includes the reference) and the target document (containing the external content).

When reusing document content, it's important to keep in mind that the Technical Document Type of the document including the content must be valid. This means that the root nodes in the document to include must be able to exist at the location in the source document that they will be placed. For example, if a *Table* was going to include external content for its rows, each document that is included must only contains *Row* Document Elements as its top-most Document Elements. For each Technical Document, the system will store the name of the top-most (root) node in the document if there is only one. Otherwise, it will the term 'multiple'. Documents with multiple root nodes are harder to reuse, especially if the root nodes are not the same type. This is because the sequence and type of root nodes must conform to every document that they are being reused in. For this reason, it is generally beneficial to only have a single root node for any Document that the author intends to reuse. If not a single root, then have all root nodes be of the same type.

### 3.5.1 Creating a Reference to External Content

At any location where either a child or peer Document is allowed, the context menus will include a sub-menu entitled 'External Content' (Figure 15 and Figure 16). Selecting this menu item will display a search grid for Technical Documents. The searched content will be filtered by Documents that use the same Technical Document Type and that include root nodes of a type that can be added (or inserted as the case may be) where the author has chosen to add the content. The author adds whatever additional search criteria as is necessary and selects the Technical Document instance to add.

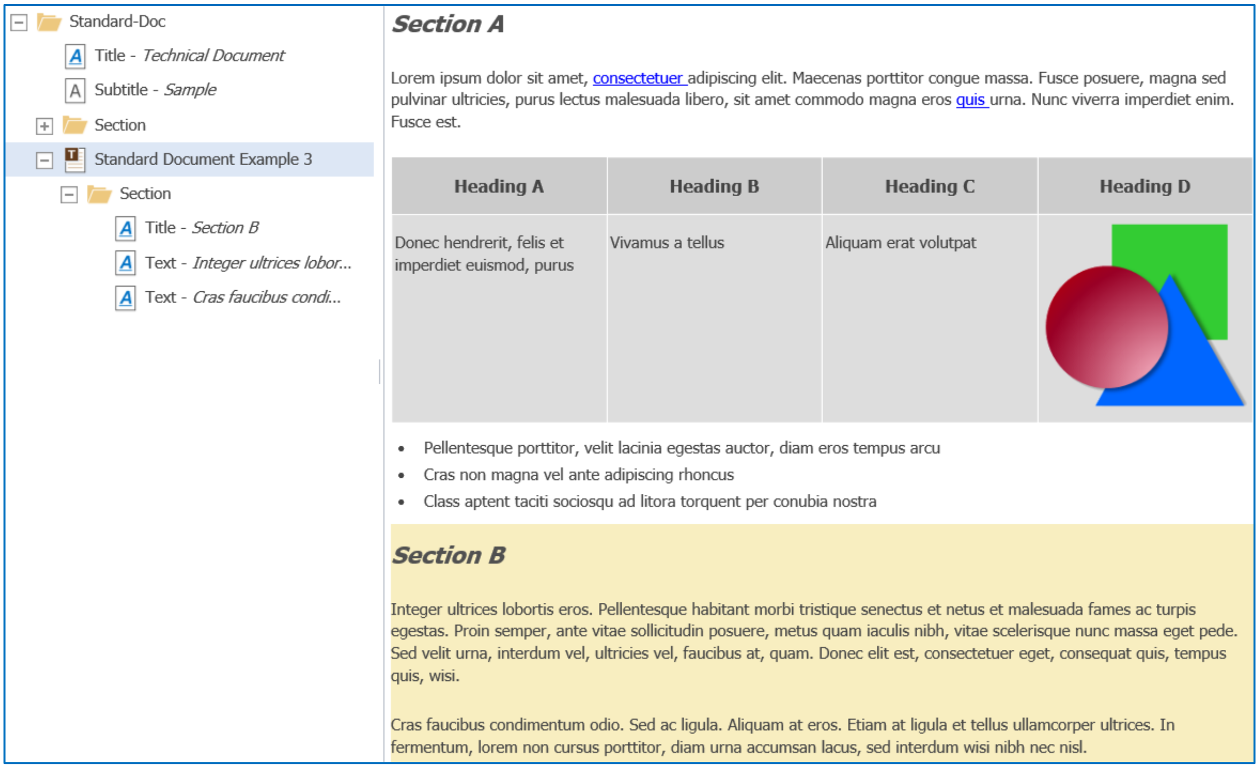



Figure 31.

External content will display in the Navigation Pane as a separate node using the name of the Technical Document that is included.  Using Figure 31 as an example, the Technical Document 'Standard Document Example 3' was inserted.  Note that the Document Element *Standard-Doc* must have a *Title* Document as the first child, an option *Subtitle* as the second child and any number of *Section* Document Elements that follow.  Since the Technical Document that was inserted had a *Section* Document Element as its root, it was a candidate for insertion at the location it was inserted.  In the Content Editor, the added external content looks identical to the content that exists within the document.

### 3.5.2 Converting External Content

Any external content that is referenced can be brought into a document as permanent content.  In this case a copy of the content of the referenced document is added to the source document, replacing the reference.  To copy the contents of a referenced document to the location where it is referenced, right-click the *External Content* Document Element in the Navigation Pane and select **Make Internal.**

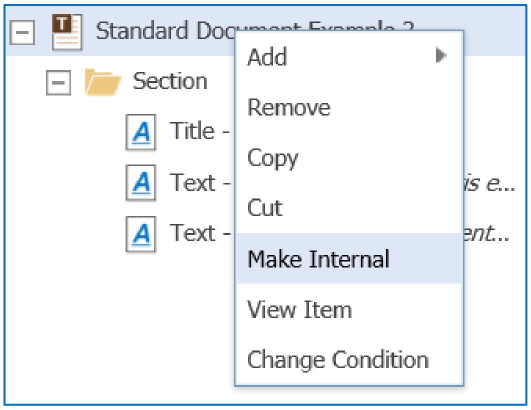



Figure 32.

The external content converted to internal is grouped using a Group Document Element.  This can then be ungrouped.  See Section 3.4.1.

### 3.5.3 Creating External Content

A separate Technical Document instance can be created from selected Document Elements.  This is useful when the author realizes that there is content within a current document that can be reused in other documents.  To create a Technical Document using selected, contiguous Document Elements select the Document Element(s) using the left mouse button.  To select multiple Document Elements, hold down the **Control** Key while selecting each Document Element.  Right-click on one of the selected Document Elements and select **Make External**.  The following dialog will be displayed:

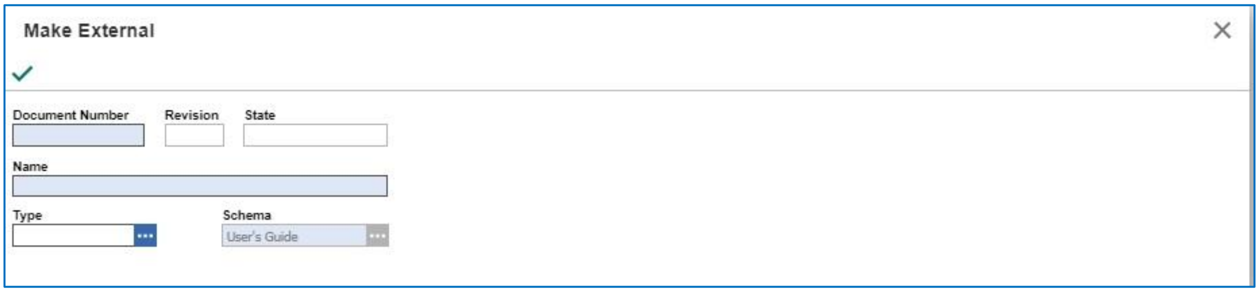



Figure 33.

Provide the Name and Document Number for the Technical Document and select **OK**. After selecting OK, save the original document at which point the external Technical Document will be created and the external reference between the two documents created.

## 3.6 Storing and Viewing Content in Separate Languages

The Technical Documentation Application has provisions to support storage and management of multiple language instances of document content. The content for each supported language evolves with the original. The assumed process is that a Technical Document is authored in one language (the default language) with 0 or more additional translated versions of this content stored with the document. Translated content is immutable; only the original default content will be edited by the author. The concept is shown below:

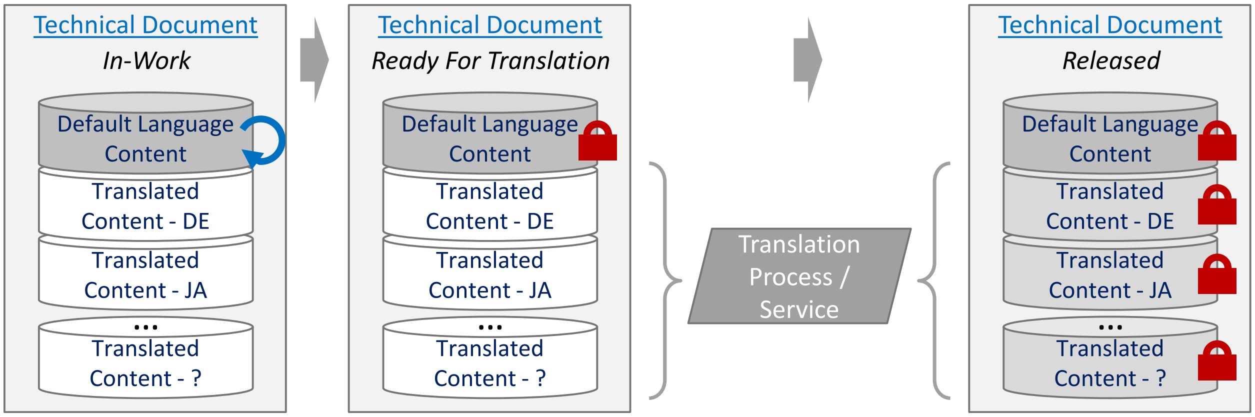



Figure 34.

Referring to Figure 33, authors will create and update document content as part of their organization's normal workflow. At some point the document content is deemed complete and hence ready to be translated. The Technical Documentation application does not have the capability to translate between languages. This must be done by an external process or service. There is also no ability for the Technical Documentation application to verify the correctness or completeness of translated content. Once the translated content is complete, it is stored with the original Technical Document in language-specific properties.

**Note:** It is important to note that document content is stored as XML with all the necessary markup that enables this content to be managed by the Technical Documentation Editor. All text is embedded within this markup. All translated content must include the original markup, exactly as it was stored, along with the translated text. To see the XML content, select the **Display As XML** button in the Document View Dialog (see Figure 5).

Support for additional languages and locales are configured as part of standard Innovator configuration. Note that the language and locale settings for the user's browser also affect how content is edited and displayed. Additional language names will be included in the language selection combo-box list in the toolbar. The content of the displayed language is loaded into the editor. Note that only the default language is editable; all others are treated as read-only.

## 3.7 Automated Content Generation using Methods

One of the main features of the Technical Documentation Application is its ability to reference Business Objects (e.g., Parts, Requirements, Processes, etc.) stored in Innovator and include this information within a Technical Document. These Business Objects (Items) are tied directly to each Technical Document that refers to them using standard Innovator Relationships. This binding allows users to see which Business Objects are related to each Technical Document and which Technical Documents relate to each Business Object using the Structure Browser. Technical Documents store the revision of each Business Object they are bound to so that when Business Objects evolve to subsequent revisions, authors can be notified and there's visual indications provided by the Editor when bound items have changed (see Section 5.1). In addition to this reference, property data from these Business Objects (and/or related Items) can be queried and used as Technical Document content as information in tables, lists or however is needed. Including information in this manner requires a Content Generator.

Custom content generation refers to the process of using configured Methods to generate XML content to be used within Technical Documents. The intent is to provide a mechanism to reference content from existing Items within Innovator or produce content as the result of a programmatic algorithm (or both) and provide this data in a format suitable for a Technical Document. Note that the term 'suitable' refers to content that respects the underlying document data model (schema).

Content Generators (the Methods that produce document content) can be used to create and store data statically within the document or dynamically with the content stored temporarily. In the former scenario, the Content Generator acts like a macro, automating the production of content. In the latter scenario, the Content Generator acts more like a reporting engine with the resulting content representing the most up-to-date information. Also, note in the latter scenario the generated data is transient; existing only until the content is regenerated, the document is published, or the document is closed in the Technical Document Editor.

| | |
|---|---|
| **Warning** | Authors should be mindful of exposing protected data. Content placed into the Technical Document will be exposed to anyone viewing this document in the future. |

Content Generators are triggered by the existence of a Document Element in the document that has been mapped to the Content Generator Method in the Document Configuration. Referring to Figure 13, the ItemInfo Document Element instance has been mapped to a Part. The Content Generator queries this part for various properties, which are then displayed in a Table Document Element.

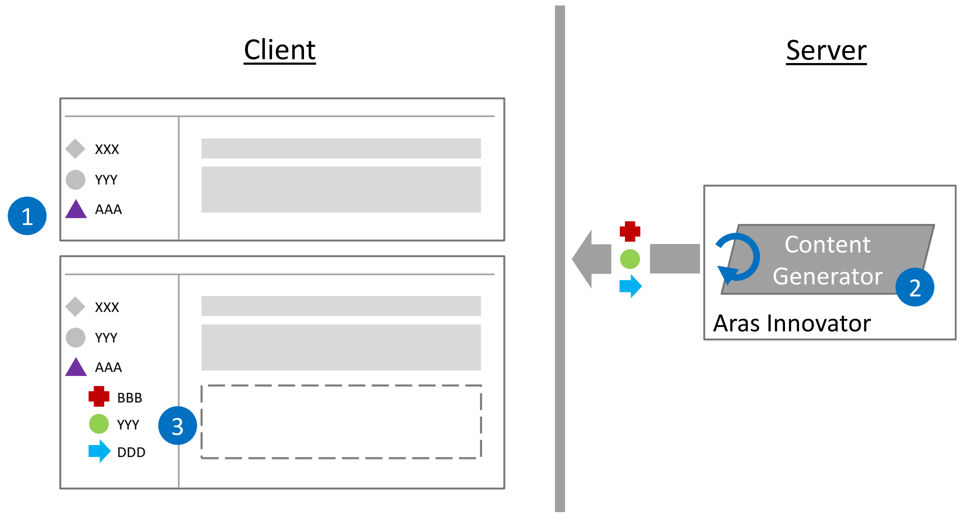



Figure 35.

The process works as follows:

1. The author places a Document Element within the document that has been mapped to a Content Generator
2. If this is an *Item* Document Element, the user will be prompted to select an Item to reference.
3. The configured Content Generator Method is called on the server to generate the Document Elements to add (as children) to the Document Element added by the author
4. The generated content is stored in the Document on the client
5. When the author saves the Technical Document, the generated content is added to the data on the server. Note that the generated content exists only on the client until it is saved.

Generated content is treated like other content created directly by the author. It can be copied and reused in other places within the document and within other Technical Documents, it can be modified, and removed. Note, however that generated content can be regenerated. Doing so will overwrite all content modified by the author.

# 4 Publishing / Exporting

The Technical Documentation application provides the capability to export the contents of a Technical Document as XML, HTML, or PDF.  This exported content can then support data exchange with customers, provide content for a website, or be the source for printed media.  HTML and PDF versions can have their own independent styling, apply the styles used by the editor, or anything in between.

**Note:** The term 'Publishing' is used synonymously with 'Export' in this document.

The Technical Documentation application provides the ability to export / publish Technical Documents and is invoked via an Action assigned to the Technical Document.  The export process executes on the server and provides the converted content to the user as a single file or a compressed file (zip file) containing multiple files.  Once the data is exported, the exported content is no longer under the control of Aras Innovator.  Managing versions of exported content is the responsibility of the end user.

The content included in an export is based on a selected Technical Document instance, a language (Section 3.6), filters (Section 3.3), linked documents (Section 3.2.6), referenced documents (Section 3.5), graphics (Section 3.2.8), and generated content (Section 3.7).  For the most part, any content displayed in the Editor can be included in an export.

**Note:** The Technical Document Editor uses specific styling and added visual indicators to support the editing process.  Some of this styling is only used within the Content Viewer.

## 4.1 Exporting a Technical Document

To Export a Technical Document, select the 'Publish Document' Action from the Actions menu while a Technical Document is opened in the editor or when a Technical Document Item is selected in the search grid. Doing so will display the Publishing Dialog

| **Note:** | It is important to save a Technical Document (if it is open in the Editor) before a document is published to update the server with the latest changes. |
|---|---|

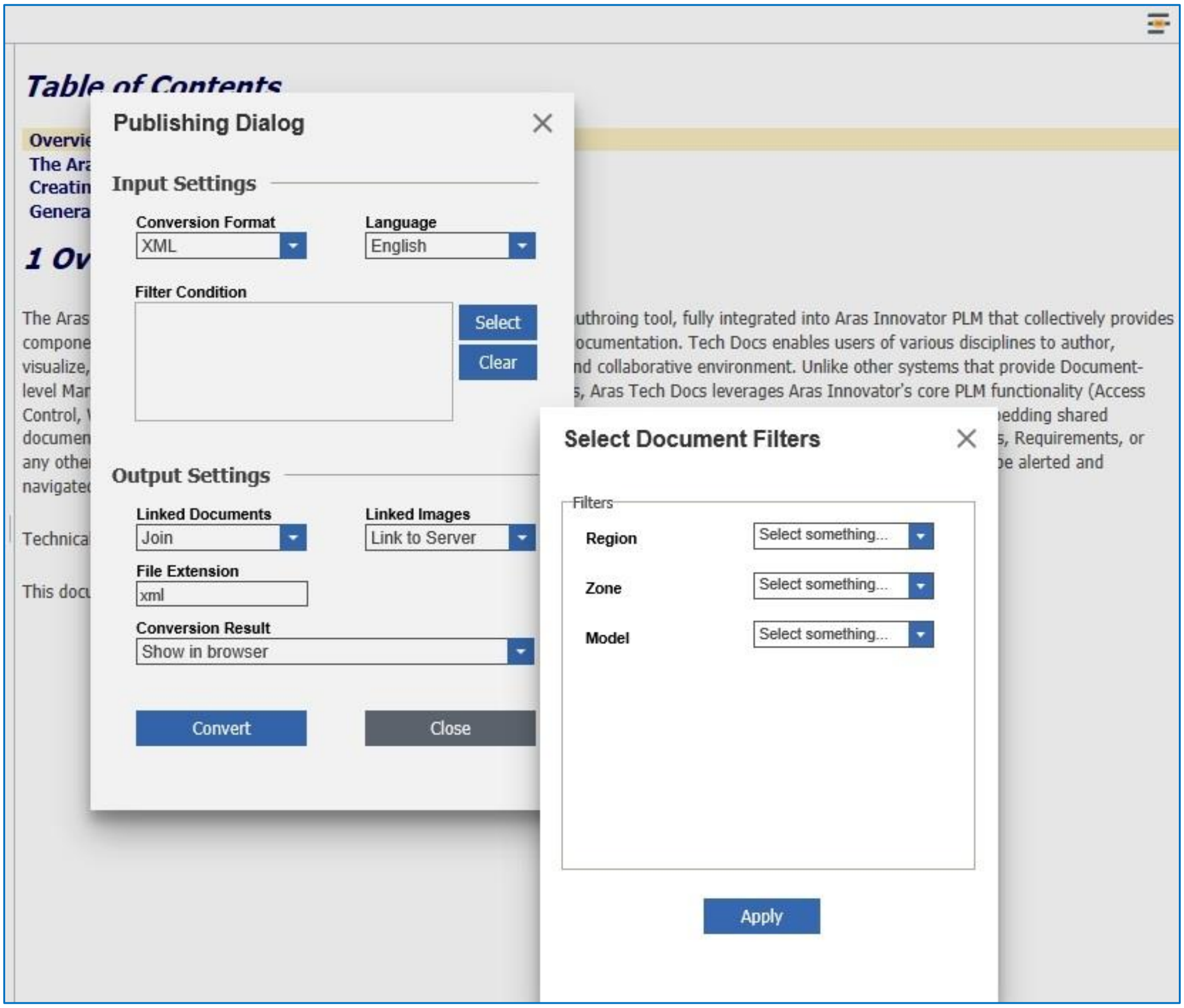



Figure 36.

The Publishing / Export process executes on the server based on the input settings provided by the user and is initiated by selecting the **Convert** button in the Publishing dialog. During conversion the user is presented with a progress dialog that is displayed until the process has completed. The execution time varies with the size of the content to convert and the bandwidth/load on the network/server at the time. Once complete, the generated content is available to the user to either display within a browser window or download to store on the user's system.

### 4.1.1 Conversion Formats

The Publish/Export function supports conversion to XML, HTML, and PDF.

#### *4.1.1.1 XML*

The content of a Technical Document is stored natively as XML and based on the Document Type it was created with. The Technical Documentation Application adds to this XML to include specific markup, most of which is then removed in the export XML content. The markup that is exported with the data includes:

- <emph> tags surrounding all formatted text
- Metadata for table and list formatting
- URLs for images
- XML elements for each Technical Document content (Block)
- Unique ids for external referenced content (images, referenced documents, Business Objects, etc.)

XML data includes just the text-based content. Images (graphics) include a URL back to the Innovator vault server for the actual image file.

#### *4.1.1.2 HTML*

The content of a Technical Document is stored natively as XML and based on the Technical Document Type it was created with. To display this content within the Editor, and when this content is exported as HTML, each Document Element is mapped to a <div> HTML tag with a class name equal to the name of the Document Element. In this way, document style rules (CSS) can target the specific elements in the HTML. All formatted text is surrounding by a <span>tag with a class name 'emph'. In addition, all style rules are included with the generated HTML within the 'Head' portion of the HTML document. All Table Document Elements are converted to <table> HTML elements and all List Document Elements are converted to <ul> or <ol> HTML elements based on the List configuration (see Section 3.2.4). Finally, all style (CSS) is included with the HTML content.

#### *4.1.1.3 PDF*

To generate PDF content, a Technical Document is first converted to HTML and then to PDF using separate server-side utility. Allstyle rules that can be applied to HTML can also be applied to the PDF content. However, PDF is a page-based media, and all PDF content is converted to a specific page size. The following additional style rules can be applied for PDF exports:

- Page numbers
- Page headers and footers
- Page margins
- Table of Content generation
- PDF Metadata

Use the following procedure to publish your document to PDF.

1. Select **Actions>Publish Document**. If you have not saved the document, the following message appears:

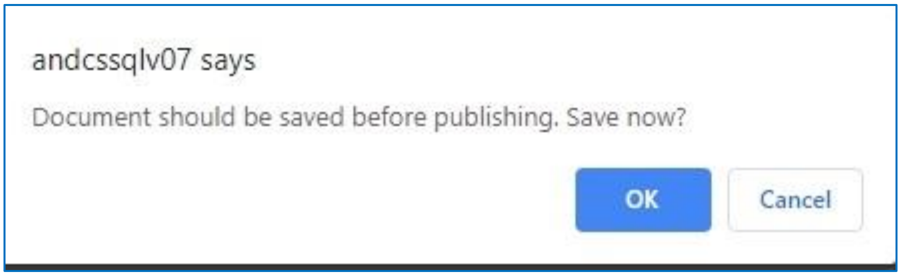



Figure 37.

2. Click **OK**. The Publishing dialog box appears.

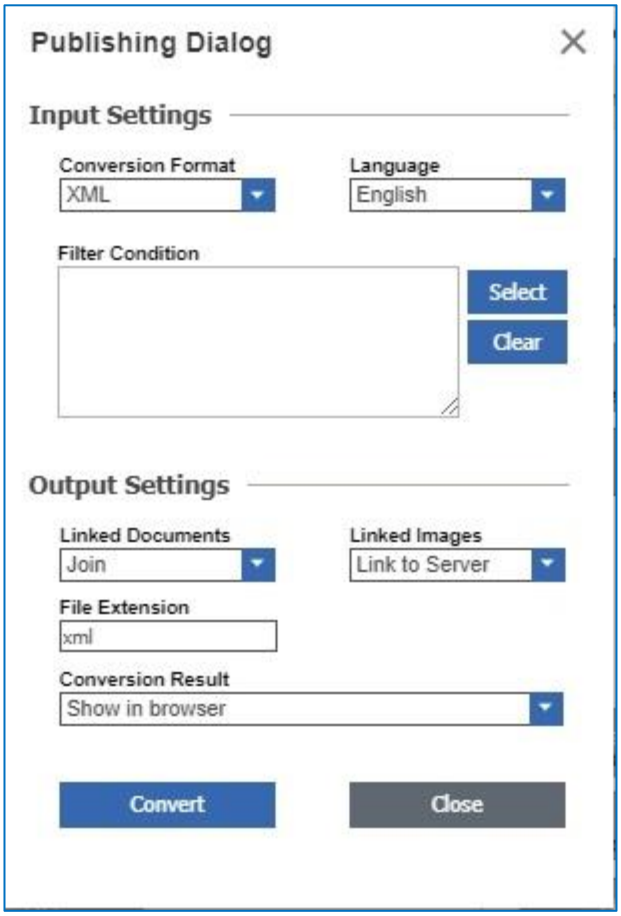



Figure 38.

3. Select **PDF** from the Conversion Format dropdown list. The File Extension value changes accordingly.

4. Click **Convert** to start the conversion process. The following message appears when the conversion process starts:

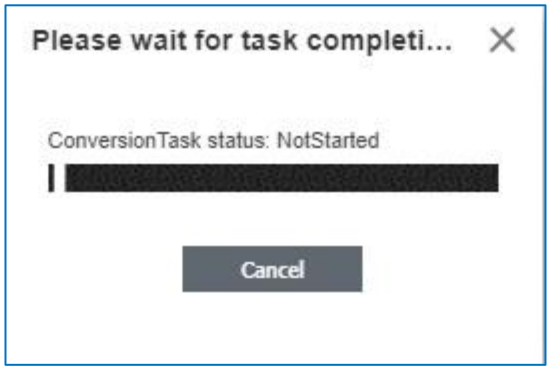
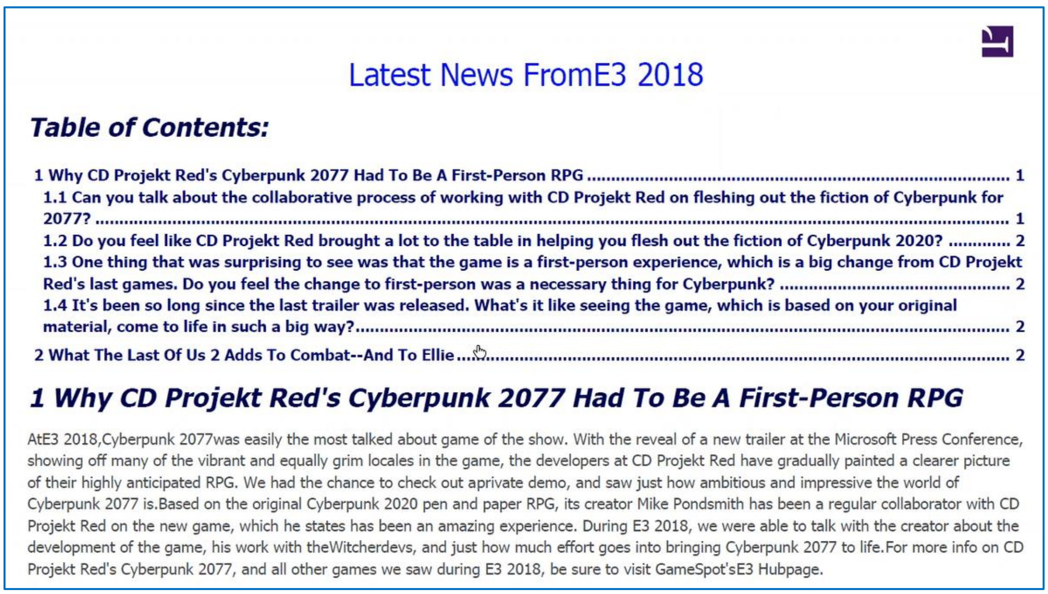



Figure 39.

When the conversion is complete, the document appears in a separate window:



Latest News FromE3 2018

**Table of Contents:**



**1 Why CD Projekt Red's Cyberpunk 2077 Had To Be A First-Person RPG**

AtE3 2018,Cyberpunk 2077was easily the most talked about game of the show. With the reveal of a new trailer at the Microsoft Press Conference, showing off many of the vibrant and equally grim locales in the game, the developers at CD Projekt Red have gradually painted a clearer picture of their highly anticipated RPG. We had the chance to check out aprivate demo, and saw just how ambitious and impressive the world of Cyberpunk 2077 is.Based on the original Cyberpunk 2020 pen and paper RPG, its creator Mike Pondsmith has been a regular collaborator with CD Projekt Red on the new game, which he states has been an amazing experience. During E3 2018, we were able to talk with the creator about the development of the game, his work with theWitcherdevs, and just how much effort goes into bringing Cyberpunk 2077 to life.For more info on CD Projekt Red's Cyberpunk 2077, and all other games we saw during E3 2018, be sure to visit GameSpot'sE3 Hubpage.

Figure 40.

The entries in the Table of Contents are links to the different sections that enable you to navigate through the document. Section and page numbers appear in the Table of Contents.

### 4.1.2 Language

The **Language** combo-box list shows each of the languages configured within Innovator.  Select the appropriate language to include in the export.  Note that the language currently selected may not be the language displayed in the Editor.

### 4.1.3 Filters

Filters can be applied the same way they are in the editor (see Section 3.3) with the same result on the content included. Note that content filtered out will be excluded from the exported content. To apply Filters, select the **Create** button next to the Filter Condition edit box in the Publishing dialog. Select the Filters to apply and select **Apply**. The selected Filters will be displayed in the Filter Condition edit box.

### 4.1.4 File Extension

A default file extension is created based on the Conversion Format chosen. Users can overwrite this value if needed and the extension string is used for all non-image files created by the export process.

### 4.1.5 Linked Content

Linked content refers to Technical Documents that are the target of a Link (see Section 3.2.6). When exporting, the user has the option to include linked content or ignore it. When including linked content, the user can choose to either embed the linked data at the end of the exported content or create a separate document and link to it. Figure 26 describes the resulting files that get created when a Technical Document that has a linked Technical Document and a Graphic included is exported.

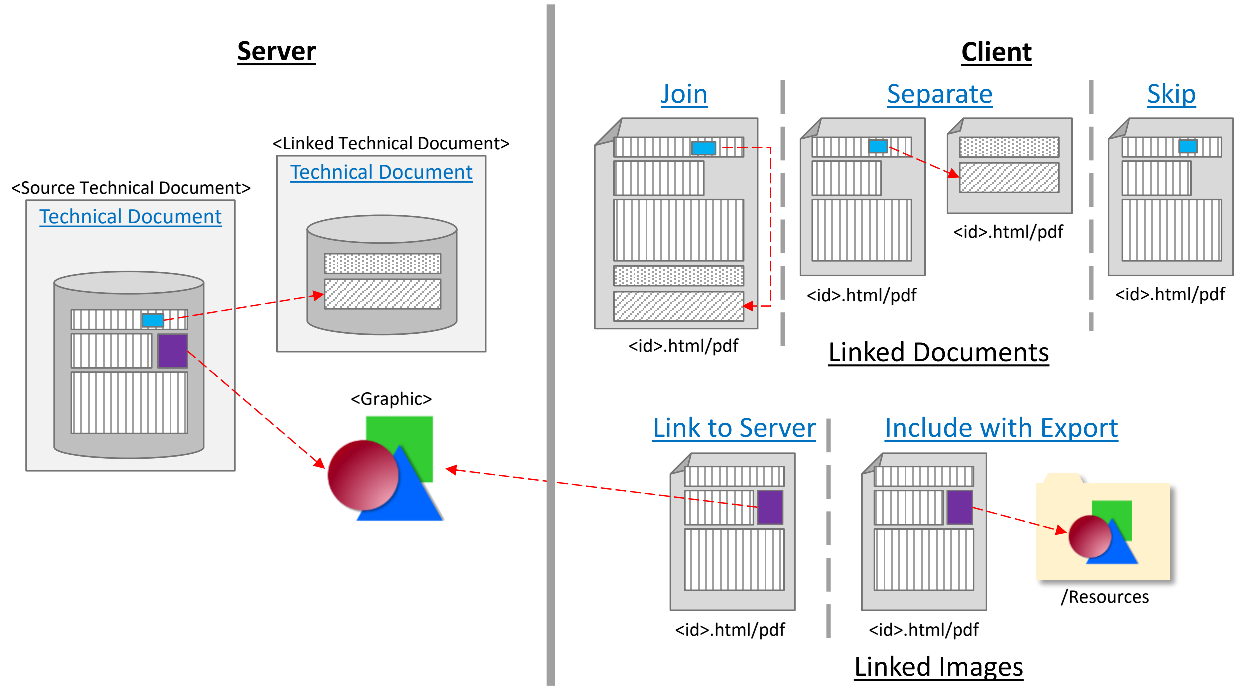



Figure 41.

#### *4.1.5.1 Linked Documents*

Linked Documents refer to the Technical Document instances that include a Document Element that is the target of a link that exists in a separate, source document. When exporting, the following options are available:

**Join:** The default. The content of the Technical Document that includes the linked Document Element is appended to the source document and the link is updated to *point* to the internal location.

**Separate:** A separate document is created, and the link is update to *point* to the generated file.

**Skip:** The content of the linked document is not included in the export

#### *4.1.5.2 Linked Images*

Linked images refer to any graphic that is included in a Technical Document. Image files are stored in the Innovator vault and accessible via URL to the vault server. When exporting, the following options are available:

**Link to Server:** The default. A URL to the vault server is included as the 'src' location of the image. Generated files refer to the innovator server for the image content.

**Include with Export:** All referenced image files are included with the export and stored in a generated sub-directory – '/Resources'. Image files are named using their ID followed by the appropriate file extension. The 'src' location of the image is updated to point to the relative path and file.

### 4.1.6 Conversion Result

Users can either view the converted results in a browser window or download and save the converted data locally on the user's file system. Note that when the user chooses to *Separate* linked documents (Section 4.1.5.1) and/or *Include* linked graphics (Section 4.1.5.2) the downloaded file will be a zip archive file containing all of the relevant related files that have been created.

**Note:** All converted data is created and stored temporarily as Files in Aras Innovator. Eventually, these files will be removed in the process of normal garbage collection.

#### *4.1.6.1 Viewing Converted Data in the Browser*

To view converted data in a browser window, select **Show in browser** from the Conversion Result combo-box list. After the conversion process has completed, a separate browser window will be opened to display the converted data. In this case, the converted data is left on the server and the browser displays it.

#### *4.1.6.2 Downloading Converted Files*

To download and save converted content, select **Save into file** from the Conversion Result combo-box list. After the conversion process has completed, the user will be prompted to identify the location to save the converted results.

# 5 Managing Technical Documents

The functionality provided by the Technical Documentation application, along with the core capabilities of Aras Innovator, enable an organization to distribute technical documentation content authoring across the enterprise.  The enabling features include Versioning, Access Control, Relationships, Lifecycle, and Workflow.  This section describes how change is managed using Versioning and identifies the default Permissions and Lifecycle Maps for Access Control and Lifecycles respectively.

## 5.1 Change Management

A Technical Document is seldom a single, self-contained Item.  It is expected that each will frequently contain other referenced document material, graphics, and information from related Business Objects (Items).  All these components evolve independently and thus affect the documents that refer to them. Relationship configuration between Technical Documents and all referenced Items will control how documents are updated.  Figure 42 shows the default Relationships from a Technical Document.  Note that all referenced Items are versioned, including Technical Documents.  Also note that, other than a Link Relationship, all Relationships are fixed.  That is a Technical Document will refer to a specific Version of each related Item for all content that is, or that potentially is, included in the document.

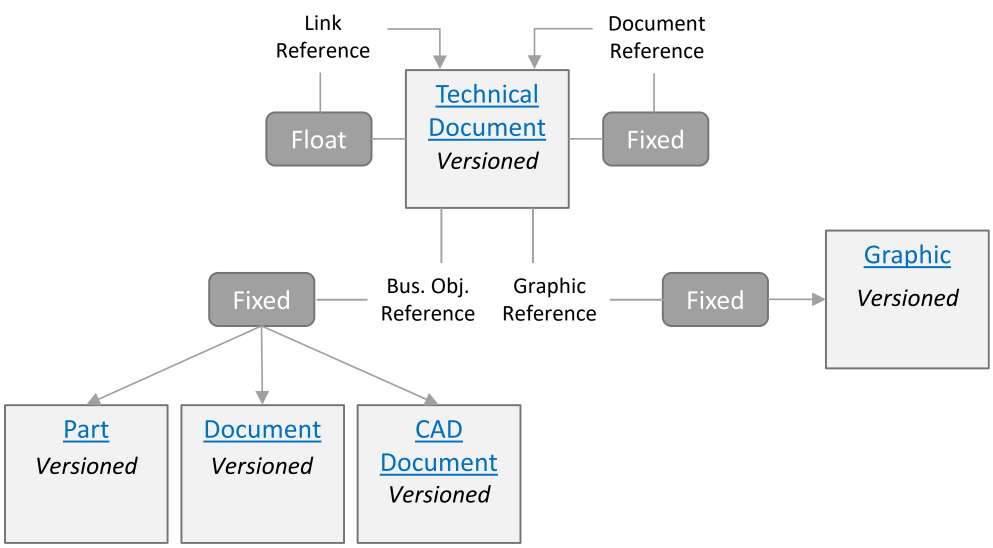



Figure 42.

### 5.1.1 Viewing Update Referenced Items

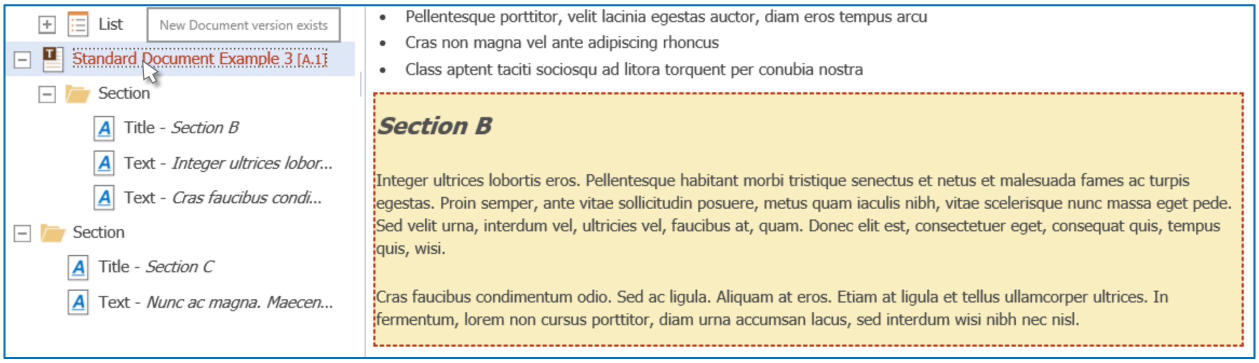



Figure 43.

When a referenced Item (Technical Document, Business Object, or Graphic) has been updated the content contained in the source document is highlighted. The node in the Navigation Pane for the top-most Document Element will be rendered in red with the current Revision and Version appended to the node text in square brackets and an added tooltip signifying the existence of the updated Item. The content for the referenced Item is surrounded in a red, dotted border. Note that the original content of the document is still displayed.

### 5.1.2 Viewing the Latest Referenced Item

An author can view the latest version of a referenced Item. Doing so will load the Form of the ItemType. For example, for referenced Technical Documents, the Technical Document Editor will be opened showing the contents of the latest version; for Graphics, the Form for the Graphic will be displayed; for referenced Business Objects, the associated Form for that Item will be displayed. Note that for referenced Business Objects that include generated content (see Section 3.7) only the Form of the referenced Item will be displayed.

Document Elements that refer to referenced data that has been updated will have the following context sub-menu - Version:

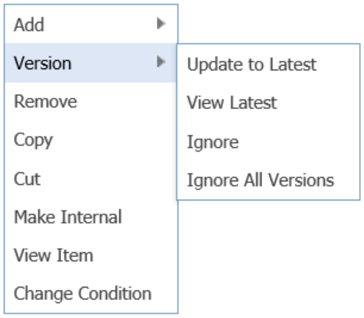



Figure 44.

Select **View Latest** to load the appropriate Form showing the latest version of the referenced Item.

### 5.1.3 Updating Referenced Items

To view the latest content from the referenced Item, select **Update to Latest** in the **Version** sub-menu. The document will be updated based on the type of referenced content. For Graphics, the latest image will be displayed. For reference Technical Documents, the latest content of the document will be displayed. For Business Objects with generated content, the Content Generator will be called on the latest version of the referenced Item and the document updated accordingly. As always, users must save the Technical Document to apply the changes to the document and update the corresponding Relationships.

### 5.1.4 Ignoring Updated Referenced Data

Authors can opt to ignore subsequent versions of referenced content either by selection **Ignore** or **Ignore All Versions** in the **Version** sub-menu. By selecting **Ignore**, only the specific latest version will be ignored. By selecting **Ignore All Versions**, all subsequent versions will be ignored. Ignoring updated versions only means that the Editor will not prompt the user with the styling applied inFigure 43. The data associated with the currently referenced Item will always be displayed. However, once the author chooses to ignore updated versions of referenced content, they can restore tracking of these versions (and thus turn the display of updated content back on) by choosing **Resume Version Tracking** in the **Version** sub-menu.

## 5.2 Access Control / Permissions

The Technical Documentation application includes to Identities to account for the two types of users:

- TECHNICAL DOCUMENT AUTHOR
- TECHNICAL DOCUMENT ADMINISTRATOR

### 5.2.1 Technical Document Author

A Technical Document Author's responsibility is to create Technical Document content with the appropriate relationships to relevant Items. They are the primary user of the Technical Document Editor (Figure 2) and referred to in this document as 'author'.

A Technical Document Author is able create, read, update, and delete Technical Documents and Graphics. They should have read access for Document Structures (and related items) as well as the ability to execute any Method configured as part of Document Type definition (i.e., Content Generators (see Section 3.7)). They should have access to all capabilities of the Technical Document Editor, including all available features of the Publishing function.

### 5.2.2 Technical Document Administrator

A Technical Document Administrator is responsible for configuring the available Document Data models (i.e., define Document Type definitions with associated Renderers, Content Generators, Schema Element configuration, and Styling). In addition to the permissions of a Technical Document Author, an administrator should be able to create, read, update, delete, discover, and modify the access for a Document Type and its related items, including any Methods provided as part of the Technical Document Type configuration.

### 5.2.3 Permissions on Technical Document and Graphics

The following describes the default Permissions for the Technical Document and Graphic Items.

| Identity | Create | Get | Update | Delete | Discover | Change Permission |
|---|---|---|---|---|---|---|
| All Employees | | X | | | X | |
| Technical Document Author | X | X | X | | X | |
| Creator | | X | X | X | X | |
| Technical Document Administrator | X | X | X | X | X | X |

By default, all employees will be able to search for, open and read a Technical Document along with its graphical content. Note that referenced data from Content Generators may not be viewable to these users depending on the access permissions of the related Items. The *Creator* Identity works in tandem with the Technical Document Author permissions in that only the Technical Document and Graphics created by a specific user can be removed by that user. Technical Document Administrators have the additional ability to change permissions for all Technical Documents and Graphics.

### 5.2.4 Blocked Content

Depending on the specific Permissions allotted to the specific user, it's possible that referenced content within a Technical Document may not be accessible. When referenced content is not accessible to a user, the Technical Document Editor will not show the content and will provide visual indications in the Navigation Pane and Content Viewer for all blocked content.

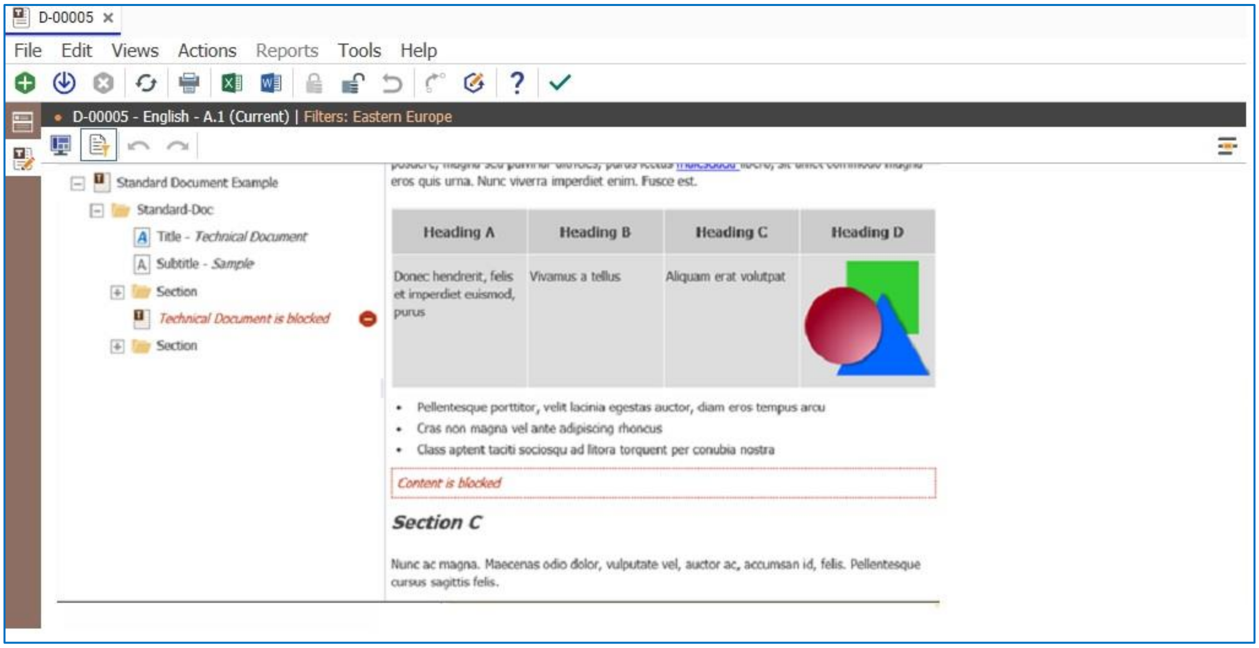



Figure 45.

All Document Elements represented as blocked referenced content will be displayed in the Navigation Pane with red italics and an added icon to the right of the node.  Any child elements will not be included in the display.  In addition, the area in the Content Viewer where the associated content would display is replaced with a single block section with a red, dotted border and the static text 'Content is blocked' displayed within it.  Document Elements that reference blocked content can be removed, however from a Technical Document.

## 5.3 Lifecycle

By default, a Technical Document will have the following lifecycle:

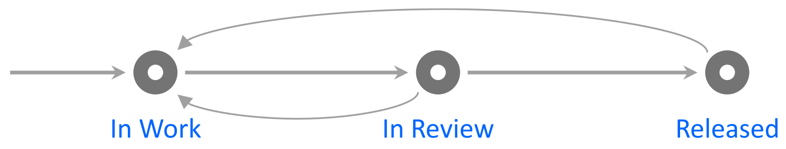



Figure 46.

The starting state is *In Work* with Permissions as shown above.  The *In Review* and *Released* states have the following Permissions:

| Identity | Create | Get | Update | Delete | Discover | Change Permission |
|---|---|---|---|---|---|---|
| All Employees | | X | | | X | |
| Technical Document Author | | X | | | X | |
| Technical Document Administrator | X | X | X | X | X | X |

Note that all states maintain a fixed relationship with related Items.  Graphics have a similar lifecycle and Permission model, but with only *In Work* and *Released* states.